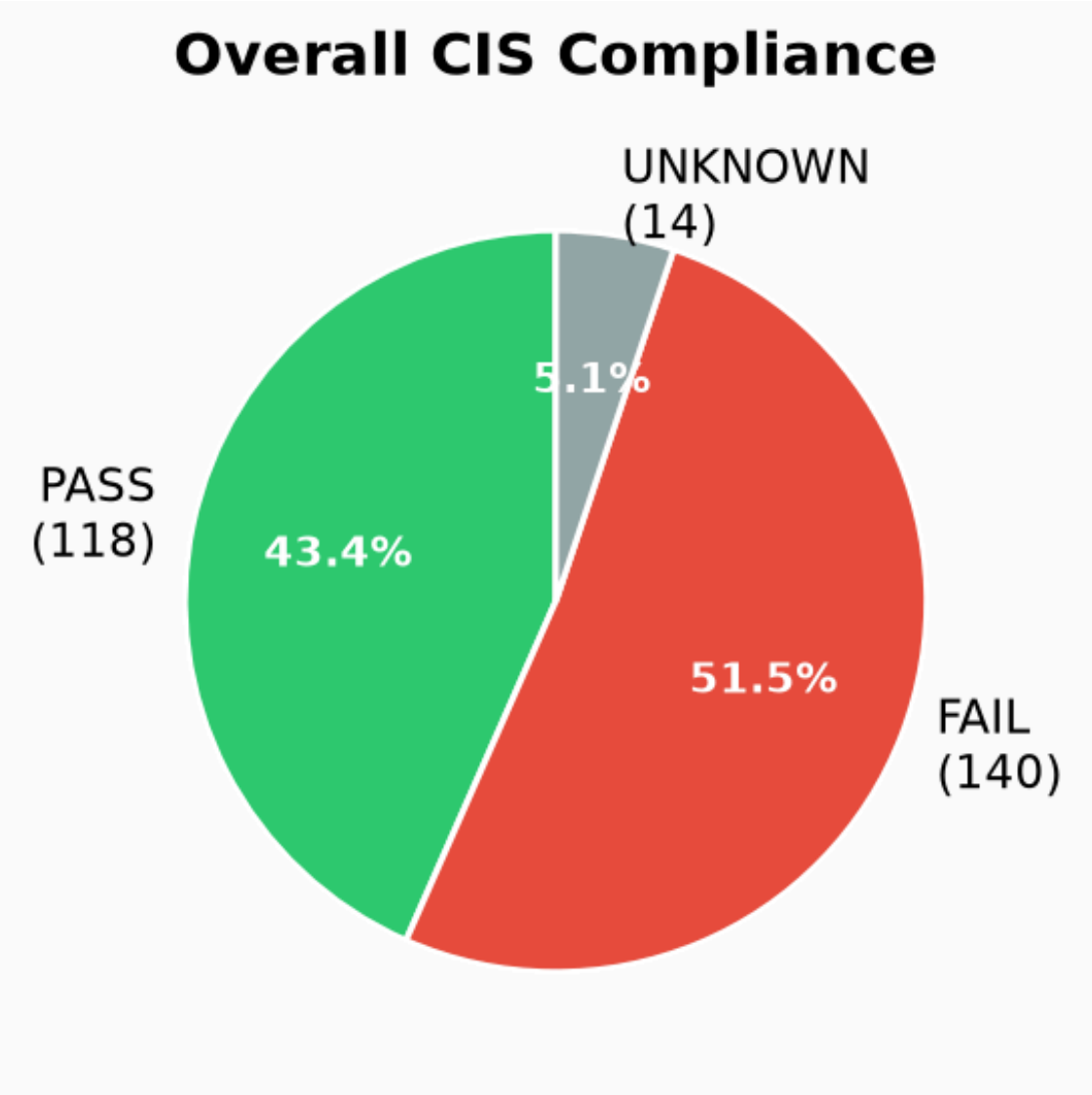


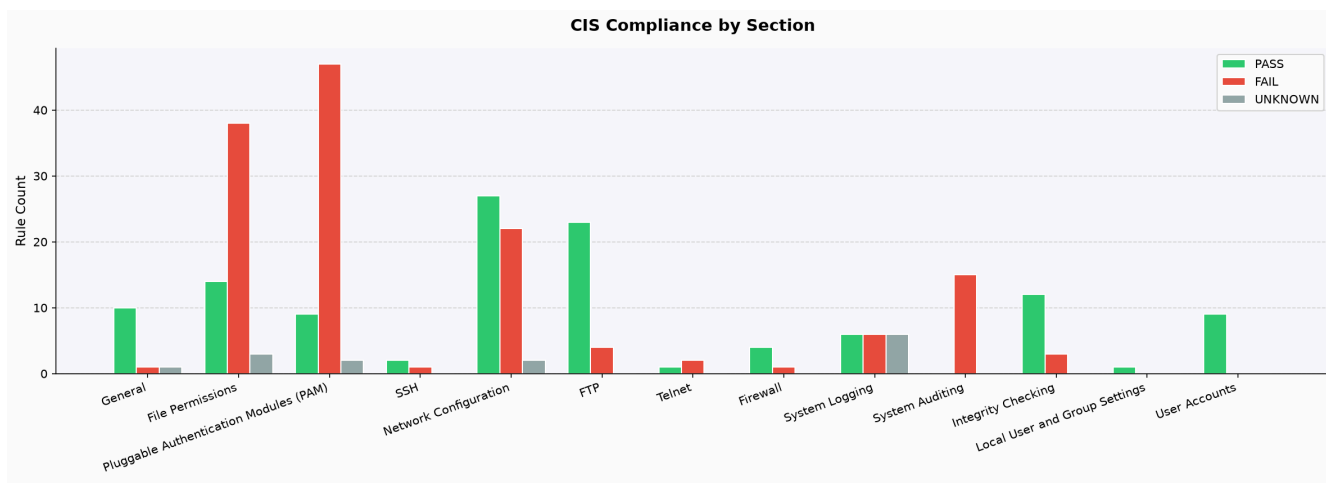
SENTINEL — CIS Benchmark Audit Report

Executive Summary

Metric	Value
Total Rules Evaluated	272
✓ Passed	118
✗ Failed	140
? Unknown	14
Overall Pass Rate	43.4%

Compliance Charts





Per-Section Breakdown

Section	✓ PASS	✗ FAIL	? UNKNOWN	Pass Rate
General	10	1	1	83%
File Permissions	14	38	3	25%
Pluggable Authentication Modules (PAM)	9	47	2	16%
SSH	2	1	0	67%
Network Configuration	27	22	2	53%
FTP	23	4	0	85%
Telnet	1	2	0	33%
Firewall	4	1	0	80%
System Logging	6	6	6	33%
System Auditing	0	15	0	0%
Integrity Checking	12	3	0	80%
Local User and Group Settings	1	0	0	100%
User Accounts	9	0	0	100%

CIS Benchmark Compliance Report

Here is the security compliance report for the remote machine based on the provided CIS Ubuntu 24.04 LTS Benchmark rules and security profile.

Section 1.1 – Filesystem Kernel Modules & Partitions

Rule 1.1.1.1: Ensure cramfs kernel module is not available

- **Status:** PASS
- **Evidence:** `cramfs` module is not currently loaded and no `modprobe` configuration lines were found (implying it's not explicitly blacklisted or disabled, but also not loaded).

○ `filesystem.filesystem_kernel_modules[0].currently_loaded:`

```
    false
  ○ filesystem.filesystem_kernel_modules[0].modprobe_config_lines:
    []
```

- **Recommendation:** None.

Rule 1.1.1.2: Ensure freevxfs kernel module is not available

- **Status:** PASS
- **Evidence:** freevxfs module is not currently loaded and no modprobe configuration lines were found.

```
  ○ filesystem.filesystem_kernel_modules[1].currently_loaded:
    false
  ○ filesystem.filesystem_kernel_modules[1].modprobe_config_lines:
    []
```

- **Recommendation:** None.

Rule 1.1.1.3: Ensure hfs kernel module is not available

- **Status:** PASS
- **Evidence:** hfs module is not currently loaded and no modprobe configuration lines were found.

```
  ○ filesystem.filesystem_kernel_modules[2].currently_loaded:
    false
  ○ filesystem.filesystem_kernel_modules[2].modprobe_config_lines:
    []
```

- **Recommendation:** None.

Rule 1.1.1.4: Ensure hfsplus kernel module is not available

- **Status:** PASS
- **Evidence:** hfsplus module is not currently loaded and no modprobe configuration lines were found.

```
  ○ filesystem.filesystem_kernel_modules[3].currently_loaded:
    false
  ○ filesystem.filesystem_kernel_modules[3].modprobe_config_lines:
    []
```

- **Recommendation:** None.

Rule 1.1.1.5: Ensure jffs2 kernel module is not available

- **Status:** PASS
- **Evidence:** jffs2 module is not currently loaded and no modprobe configuration lines were found.

```
  ○ filesystem.filesystem_kernel_modules[4].currently_loaded:
    false
  ○ filesystem.filesystem_kernel_modules[4].modprobe_config_lines:
```

[]

- **Recommendation:** None.

Rule 1.1.1.6: Ensure overlay kernel module is not available

- **Status:** PASS
- **Evidence:** overlay module is not currently loaded and no modprobe configuration lines were found.
 - `filesystem.filesystem_kernel_modules[5].currently_loaded:`
`false`
 - `filesystem.filesystem_kernel_modules[5].modprobe_config_lines:`
`[]`
- **Recommendation:** None.

Rule 1.1.1.7: Ensure squashfs kernel module is not available

- **Status:** PASS
- **Evidence:** squashfs module is not currently loaded and no modprobe configuration lines were found.
 - `filesystem.filesystem_kernel_modules[6].currently_loaded:`
`false`
 - `filesystem.filesystem_kernel_modules[6].modprobe_config_lines:`
`[]`
- **Recommendation:** None.

Rule 1.1.1.8: Ensure udf kernel module is not available

- **Status:** PASS
- **Evidence:** udf module is not currently loaded and no modprobe configuration lines were found.
 - `filesystem.filesystem_kernel_modules[7].currently_loaded:`
`false`
 - `filesystem.filesystem_kernel_modules[7].modprobe_config_lines:`
`[]`
- **Recommendation:** None.

Rule 1.1.1.9: Ensure firewire-core kernel module is not available

- **Status:** PASS
- **Evidence:** firewire-core module is not currently loaded and no modprobe configuration lines were found.
 - `filesystem.filesystem_kernel_modules[8].currently_loaded:`
`false`
 - `filesystem.filesystem_kernel_modules[8].modprobe_config_lines:`
`[]`

- **Recommendation:** None.

Rule 1.1.1.10: Ensure usb-storage kernel module is not available

- **Status:** PASS
- **Evidence:** usb-storage module is not currently loaded and no modprobe configuration lines were found.
 - `filesystem.filesystem_kernel_modules[9].currently_loaded:`
`false`
 - `filesystem.filesystem_kernel_modules[9].modprobe_config_lines:`
`[]`
- **Recommendation:** None.

Rule 1.1.1.11: Ensure unused filesystems kernel modules are not available

- **Status:** INFORMATIONAL
- **Evidence:** The following kernel modules are currently loaded on the system:
`tcp_diag, udp_diag, inet_diag, nf_tables, tls, qrtr, cfg80211, binfmt_misc, snd_intel8x0, intel_rapl_msr, snd_ac97_codec, ac97_bus, snd_pcm, intel_rapl_common, vmwgfx, snd_timer, drm_ttm_helper, snd, i2c_piix4, vboxguest, soundcore, ttm, joydev, input_leds, mac_hid, serio_raw, sch_fq_codel, dm_multipath, msr, efi_pstore, nfnetlink, dmi_sysfs, ip_tables, x_tables, autofs4, btrfs, blake2b_generic, raid10, raid456, async_raid6_recov, async_memcpy, async_pq, async_xor, async_tx, xor, raid6_pq, libcrc32c, raid1, raid0, crct10dif_pclmul, crc32_pclmul, polyval_clmulni, polyval_generic, ghash_clmulni_intel, sha256_ssse3, sha1_ssse3, vga16fb, vgastate, psmouse, ahci, e1000, libahci, pata_acpi, video, wmi, hid_generic, usbhid, hid, aesni_intel, crypto_simd, cryptd.`
- **Recommendation:** Review the list of loaded kernel modules to ensure only necessary filesystems are available. If any of these are not required, consider blacklisting them via `/etc/modprobe.d/`.

Rule 1.1.2.1.1: Ensure /tmp is tmpfs or a separate partition

- **Status:** FAIL
- **Evidence:** The `/tmp` directory is not a separate partition and is not mounted as tmpfs.
 - `filesystem.filesystem_partitions[0].mount_point:` `/tmp`
 - `filesystem.filesystem_partitions[0].is_separate_partition:`
`false`
 - `filesystem.filesystem_partitions[0].fstype:` `ext4`
- **Recommendation:** Configure `/tmp` as a separate partition or mount it as a tmpfs filesystem.
 - **To use tmpfs:** Add `tmpfs /tmp tmpfs defaults,rw,nosuid,nodev,noexec,relatime 0 0` to `/etc/fstab`.
 - **To use a separate partition:** Create a new partition, format it, and add an entry to `/etc/fstab` with appropriate options.

Rule 1.1.2.1.2: Ensure nodev option set on /tmp partition

- **Status:** FAIL
- **Evidence:** The /tmp partition is missing the nodev mount option.
 - `filesystem.filesystem_partitions[0].mount_point: /tmp`
 - `filesystem.filesystem_partitions[0].mount_options_list: ['rw', 'relatime']`
- **Recommendation:** Add the nodev option to the /tmp mount point in /etc/fstab. This prevents character or block special devices from being used on the partition.

Rule 1.1.2.1.3: Ensure nosuid option set on /tmp partition

- **Status:** FAIL
- **Evidence:** The /tmp partition is missing the nosuid mount option.
 - `filesystem.filesystem_partitions[0].mount_point: /tmp`
 - `filesystem.filesystem_partitions[0].mount_options_list: ['rw', 'relatime']`
- **Recommendation:** Add the nosuid option to the /tmp mount point in /etc/fstab. This prevents setuid/setgid bits from having any effect on the partition.

Rule 1.1.2.1.4: Ensure noexec option set on /tmp partition

- **Status:** FAIL
- **Evidence:** The /tmp partition is missing the noexec mount option.
 - `filesystem.filesystem_partitions[0].mount_point: /tmp`
 - `filesystem.filesystem_partitions[0].mount_options_list: ['rw', 'relatime']`
- **Recommendation:** Add the noexec option to the /tmp mount point in /etc/fstab. This prevents direct execution of binaries on the partition.

Rule 1.1.2.2.1: Ensure /dev/shm is tmpfs or a separate partition

- **Status:** PASS
- **Evidence:** /dev/shm is mounted as tmpfs.
 - `filesystem.filesystem_partitions[1].mount_point: /dev/shm`
 - `filesystem.filesystem_partitions[1].is_separate_partition: true`
 - `filesystem.filesystem_partitions[1].fstype: tmpfs`
- **Recommendation:** None.

Rule 1.1.2.2.2: Ensure nodev option set on /dev/shm partition

- **Status:** PASS
- **Evidence:** The /dev/shm partition has the nodev mount option set.
 - `filesystem.filesystem_partitions[1].mount_point: /dev/shm`

- `filesystem.filesystem_partitions[1].mount_options_list: ['rw', 'nosuid', 'nodev', 'inode64']`

- **Recommendation:** None.

Rule 1.1.2.2.3: Ensure nosuid option set on /dev/shm partition

- **Status:** PASS
- **Evidence:** The /dev/shm partition has the nosuid mount option set.

- `filesystem.filesystem_partitions[1].mount_point: /dev/shm`
- `filesystem.filesystem_partitions[1].mount_options_list: ['rw', 'nosuid', 'nodev', 'inode64']`

- **Recommendation:** None.

Rule 1.1.2.2.4: Ensure noexec option set on /dev/shm partition

- **Status:** FAIL
- **Evidence:** The /dev/shm partition is missing the noexec mount option.

- `filesystem.filesystem_partitions[1].mount_point: /dev/shm`
- `filesystem.filesystem_partitions[1].mount_options_list: ['rw', 'nosuid', 'nodev', 'inode64']`

- **Recommendation:** Add the noexec option to the /dev/shm mount point in /etc/fstab. This prevents direct execution of binaries on the partition.

Rule 1.1.2.3.1: Ensure separate partition exists for /home

- **Status:** FAIL
- **Evidence:** The /home directory is not a separate partition.

- `filesystem.filesystem_partitions[2].mount_point: /home`
- `filesystem.filesystem_partitions[2].is_separate_partition: false`
- `filesystem.filesystem_partitions[2].fstype: ext4`

- **Recommendation:** Configure /home as a separate partition. This isolates user data from the operating system and prevents a full /home partition from impacting system stability.

Rule 1.1.2.3.2: Ensure nodev option set on /home partition

- **Status:** FAIL
- **Evidence:** The /home partition is missing the nodev mount option.

- `filesystem.filesystem_partitions[2].mount_point: /home`
- `filesystem.filesystem_partitions[2].mount_options_list: ['rw', 'relatime']`

- **Recommendation:** Add the nodev option to the /home mount point in /etc/fstab. This prevents character or block special devices from being used on the partition.

Rule 1.1.2.3.3: Ensure nosuid option set on /home partition

- **Status:** FAIL
- **Evidence:** The /home partition is missing the nosuid mount option.
 - `filesystem.filesystem_partitions[2].mount_point: /home`
 - `filesystem.filesystem_partitions[2].mount_options_list: ['rw', 'relatime']`
- **Recommendation:** Add the nosuid option to the /home mount point in /etc/fstab. This prevents setuid/setgid bits from having any effect on the partition.

Rule 1.1.2.4.1: Ensure separate partition exists for /var

- **Status:** FAIL
- **Evidence:** The /var directory is not a separate partition.
 - `filesystem.filesystem_partitions[3].mount_point: /var`
 - `filesystem.filesystem_partitions[3].is_separate_partition: false`
 - `filesystem.filesystem_partitions[3].fstype: ext4`
- **Recommendation:** Configure /var as a separate partition. This isolates variable data (logs, spool files, temporary files) from the operating system.

Rule 1.1.2.4.2: Ensure nodev option set on /var partition

- **Status:** FAIL
- **Evidence:** The /var partition is missing the nodev mount option.
 - `filesystem.filesystem_partitions[3].mount_point: /var`
 - `filesystem.filesystem_partitions[3].mount_options_list: ['rw', 'relatime']`
- **Recommendation:** Add the nodev option to the /var mount point in /etc/fstab. This prevents character or block special devices from being used on the partition.

Rule 1.1.2.4.3: Ensure nosuid option set on /var partition

- **Status:** FAIL
- **Evidence:** The /var partition is missing the nosuid mount option.
 - `filesystem.filesystem_partitions[3].mount_point: /var`
 - `filesystem.filesystem_partitions[3].mount_options_list: ['rw', 'relatime']`
- **Recommendation:** Add the nosuid option to the /var mount point in /etc/fstab. This prevents setuid/setgid bits from having any effect on the partition.

Rule 1.1.2.5.1: Ensure separate partition exists for /var/tmp

- **Status:** FAIL
- **Evidence:** The /var/tmp directory is not a separate partition.

- `filesystem.filesystem_partitions[4].mount_point: /var/tmp`
- `filesystem.filesystem_partitions[4].is_separate_partition: false`
- `filesystem.filesystem_partitions[4].fstype: ext4`

- **Recommendation:** Configure `/var/tmp` as a separate partition. This isolates temporary files from the operating system.

Rule 1.1.2.5.2: Ensure `nodev` option set on `/var/tmp` partition

- **Status:** FAIL
- **Evidence:** The `/var/tmp` partition is missing the `nodev` mount option.

- `filesystem.filesystem_partitions[4].mount_point: /var/tmp`
- `filesystem.filesystem_partitions[4].mount_options_list: ['rw', 'relatime']`

- **Recommendation:** Add the `nodev` option to the `/var/tmp` mount point in `/etc/fstab`. This prevents character or block special devices from being used on the partition.

Rule 1.1.2.5.3: Ensure `nosuid` option set on `/var/tmp` partition

- **Status:** FAIL
- **Evidence:** The `/var/tmp` partition is missing the `nosuid` mount option.

- `filesystem.filesystem_partitions[4].mount_point: /var/tmp`
- `filesystem.filesystem_partitions[4].mount_options_list: ['rw', 'relatime']`

- **Recommendation:** Add the `nosuid` option to the `/var/tmp` mount point in `/etc/fstab`. This prevents `setuid`/`setgid` bits from having any effect on the partition.

Rule 1.1.2.5.4: Ensure `noexec` option set on `/var/tmp` partition

- **Status:** FAIL
- **Evidence:** The `/var/tmp` partition is missing the `noexec` mount option.

- `filesystem.filesystem_partitions[4].mount_point: /var/tmp`
- `filesystem.filesystem_partitions[4].mount_options_list: ['rw', 'relatime']`

- **Recommendation:** Add the `noexec` option to the `/var/tmp` mount point in `/etc/fstab`. This prevents direct execution of binaries on the partition.

Rule 1.1.2.6.1: Ensure separate partition exists for `/var/log`

- **Status:** FAIL
- **Evidence:** The `/var/log` directory is not a separate partition.

- `filesystem.filesystem_partitions[5].mount_point: /var/log`
- `filesystem.filesystem_partitions[5].is_separate_partition: false`

- `filesystem.filesystem_partitions[5].fstype: ext4`

- **Recommendation:** Configure `/var/log` as a separate partition. This isolates log data from the operating system.

Rule 1.1.2.6.2: Ensure nodev option set on `/var/log` partition

- **Status:** FAIL
- **Evidence:** The `/var/log` partition is missing the `nodev` mount option.

- `filesystem.filesystem_partitions[5].mount_point: /var/log`
- `filesystem.filesystem_partitions[5].mount_options_list: ['rw', 'relatime']`

- **Recommendation:** Add the `nodev` option to the `/var/log` mount point in `/etc/fstab`. This prevents character or block special devices from being used on the partition.

Rule 1.1.2.6.3: Ensure nosuid option set on `/var/log` partition

- **Status:** FAIL
- **Evidence:** The `/var/log` partition is missing the `nosuid` mount option.

- `filesystem.filesystem_partitions[5].mount_point: /var/log`
- `filesystem.filesystem_partitions[5].mount_options_list: ['rw', 'relatime']`

- **Recommendation:** Add the `nosuid` option to the `/var/log` mount point in `/etc/fstab`. This prevents `setuid`/`setgid` bits from having any effect on the partition.

Rule 1.1.2.6.4: Ensure noexec option set on `/var/log` partition

- **Status:** FAIL
- **Evidence:** The `/var/log` partition is missing the `noexec` mount option.

- `filesystem.filesystem_partitions[5].mount_point: /var/log`
- `filesystem.filesystem_partitions[5].mount_options_list: ['rw', 'relatime']`

- **Recommendation:** Add the `noexec` option to the `/var/log` mount point in `/etc/fstab`. This prevents direct execution of binaries on the partition.

Rule 1.1.2.7.1: Ensure separate partition exists for `/var/log/audit`

- **Status:** FAIL
- **Evidence:** The `/var/log/audit` directory is not a separate partition.

- `filesystem.filesystem_partitions[6].mount_point: /var/log/audit`
- `filesystem.filesystem_partitions[6].is_separate_partition: false`
- `filesystem.filesystem_partitions[6].fstype: ext4`

- **Recommendation:** Configure `/var/log/audit` as a separate partition. This isolates

audit logs from other system logs and the operating system.

Rule 1.1.2.7.2: Ensure nodev option set on /var/log/audit partition

- **Status:** FAIL
- **Evidence:** The /var/log/audit partition is missing the nodev mount option.
 - `filesystem.filesystem_partitions[6].mount_point: /var/log/audit`
 - `filesystem.filesystem_partitions[6].mount_options_list: ['rw', 'relatime']`
- **Recommendation:** Add the nodev option to the /var/log/audit mount point in /etc/fstab. This prevents character or block special devices from being used on the partition.

Rule 1.1.2.7.3: Ensure nosuid option set on /var/log/audit partition

- **Status:** FAIL
- **Evidence:** The /var/log/audit partition is missing the nosuid mount option.
 - `filesystem.filesystem_partitions[6].mount_point: /var/log/audit`
 - `filesystem.filesystem_partitions[6].mount_options_list: ['rw', 'relatime']`
- **Recommendation:** Add the nosuid option to the /var/log/audit mount point in /etc/fstab. This prevents setuid/setgid bits from having any effect on the partition.

Rule 1.1.2.7.4: Ensure noexec option set on /var/log/audit partition

- **Status:** FAIL
- **Evidence:** The /var/log/audit partition is missing the noexec mount option.
 - `filesystem.filesystem_partitions[6].mount_point: /var/log/audit`
 - `filesystem.filesystem_partitions[6].mount_options_list: ['rw', 'relatime']`
- **Recommendation:** Add the noexec option to the /var/log/audit mount point in /etc/fstab. This prevents direct execution of binaries on the partition.

Section 1.2 – Package Management

Rule 1.2.1.1: Ensure the source.list and .source files use the Signed-By option

- **Status:** INFORMATIONAL
- **Evidence:**
 - `/etc/apt/sources.list`: No active repository lines found.
 - `/etc/apt/sources.list.d/ubuntu.sources`: All active repository lines use the Signed-By option.

- `/etc/apt/sources.list.d/wazuh.list`: All active repository lines use the `Signed-By` option.

- **Recommendation:** None. All active repository files are configured to use the `Signed-By` option.

Rule 1.2.1.2: Ensure weak dependencies are configured

- **Status:** FAIL
- **Evidence:** No configuration lines were found in `/etc/apt/apt.conf.d/` or `/etc/apt/apt.conf` to disable weak dependencies (`Recommends` or `Suggests`).

- `package_management.package_repositories.weak_deps_config_lines:`
`[]`

- **Recommendation:** Create or modify a file in `/etc/apt/apt.conf.d/` (e.g., `99-cis-hardening.conf`) to include:

```
APT::Install-Recommends "false";  
APT::Install-Suggests "false";
```

Rule 1.2.1.3: Ensure access to gpg key files are configured

- **Status:** PASS
- **Evidence:** All GPG key files in `/etc/apt/trusted.gpg.d/` have appropriate permissions (mode `0o644`, owner `root`, group `root`).

- `/etc/apt/trusted.gpg.d/ubuntu-keyring-2012-cdimage.gpg`: `0o644`
`root root`
- `/etc/apt/trusted.gpg.d/ubuntu-keyring-2018-archive.gpg`: `0o644`
`root root`

- **Recommendation:** None.

Rule 1.2.1.4: Ensure access to `/etc/apt/trusted.gpg.d` directory is configured

- **Status:** PASS
- **Evidence:** The `/etc/apt/trusted.gpg.d` directory has appropriate permissions (mode `0o755`, owner `root`, group `root`).

- `package_management.package_repositories.trusted_gpg_d_stat.mode`
`0o755`
- `package_management.package_repositories.trusted_gpg_d_stat.owne`
`root`
- `package_management.package_repositories.trusted_gpg_d_stat.grou`
`root`

- **Recommendation:** None.

Rule 1.2.1.5: Ensure access to `/etc/apt/auth.conf.d` directory is configured

- **Status:** PASS

- **Evidence:** The `/etc/apt/auth.conf.d` directory exists and has appropriate permissions (mode `0o755`, owner `root`, group `root`).

- `package_management.package_repositories.auth_conf_d_stat.exists`
`true`
- `package_management.package_repositories.auth_conf_d_stat.mode_c`
`0o755`
- `package_management.package_repositories.auth_conf_d_stat.owner:`
`root`
- `package_management.package_repositories.auth_conf_d_stat.group:`
`root`

- **Recommendation:** None.

Rule 1.2.1.6: Ensure access to files in the `/etc/apt/auth.conf.d/` directory is configured

- **Status:** PASS
- **Evidence:** No files were found in `/etc/apt/auth.conf.d/`.

- `package_management.package_repositories.auth_conf_d_files_stat:`
`[]`

- **Recommendation:** None. If files are added to this directory in the future, ensure they have permissions of `0o600`, owned by `root:root`.

Rule 1.2.1.7: Ensure access to `/usr/share/keyrings` directory is configured

- **Status:** PASS
- **Evidence:** The `/usr/share/keyrings` directory has appropriate permissions (mode `0o755`, owner `root`, group `root`).

- `package_management.package_repositories.keyrings_dir_stat.mode_`
`0o755`
- `package_management.package_repositories.keyrings_dir_stat.owner`
`root`
- `package_management.package_repositories.keyrings_dir_stat.group`
`root`

- **Recommendation:** None.

Rule 1.2.1.8: Ensure access to `/etc/apt/sources.list.d` directory is configured

- **Status:** PASS
- **Evidence:** The `/etc/apt/sources.list.d` directory has appropriate permissions (mode `0o755`, owner `root`, group `root`).

- `package_management.package_repositories.sources_list_d_stat.moc`
`0o755`
- `package_management.package_repositories.sources_list_d_stat.owr`
`root`
- `package_management.package_repositories.sources_list_d_stat.grc`

root

- **Recommendation:** None.

Rule 1.2.1.9: Ensure access to files in /etc/apt/sources.list.d are configured

- **Status:** PASS
- **Evidence:** All files in /etc/apt/sources.list.d/ have appropriate permissions (mode 0o644, owner root, group root).
 - /etc/apt/sources.list.d/ubuntu.sources: 0o644 root root
 - /etc/apt/sources.list.d/ubuntu.sources.curtin.orig: 0o644 root root
 - /etc/apt/sources.list.d/wazuh.list: 0o644 root root
- **Recommendation:** None.

Rule 1.2.2.1: Ensure updates, patches, and additional security software are installed

- **Status:** INFORMATIONAL
- **Evidence:** There are 85 upgradable packages on the system. The unattended-upgrades service is installed, enabled, and active, with automatic updates configured for security and ESM repositories.
 - package_management.package_updates.upgradable_package_count: 85
 - package_management.package_updates.unattended_upgrades_installed: true
 - package_management.package_updates.unattended_upgrades_service_enabled
 - package_management.package_updates.unattended_upgrades_service_active
 - package_management.package_updates.auto_upgrades_conf_content: APT::Periodic::Update-Package-Lists "1"; APT::Periodic::Unattended-Upgrade "1";
- **Recommendation:** Manually review and install the available updates using `sudo apt update && sudo apt upgrade`. Ensure that the unattended-upgrades configuration is sufficient for the system's operational requirements and security posture.

Section 1.3 – Mandatory Access Control

Rule 1.3.1.1: Ensure apparmor packages are installed

- **Status:** FAIL
- **Evidence:** The apparmor package is installed, but apparmor-utils is not.
 - apparmor.packages.apparmor_installed: true
 - apparmor.packages.apparmor_utils_installed: false

- **Recommendation:** Install the `apparmor-utils` package: `sudo apt install apparmor-utils`.

Rule 1.3.1.2: Ensure AppArmor is enabled

- **Status:** FAIL
- **Evidence:** AppArmor is loaded and its service is enabled and active, but the GRUB command line does not explicitly include `apparmor=1 security=apparmor`.
 - `apparmor.enabled_status.sysfs_enabled_value: Y`
 - `apparmor.enabled_status.service_enabled: enabled`
 - `apparmor.enabled_status.service_active: active`
 - `apparmor.enabled_status.grub_cmdline_raw: GRUB_CMDLINE_LINUX=""`
 - `apparmor.enabled_status.grub_apparmor_flag: false`
 - `apparmor.enabled_status.grub_security_flag: false`
- **Recommendation:** Edit `/etc/default/grub` and add `apparmor=1 security=apparmor` to the `GRUB_CMDLINE_LINUX` variable. Then run `sudo update-grub` and reboot the system.

Rule 1.3.1.3: Ensure all AppArmor Profiles are enforcing

- **Status:** FAIL
- **Evidence:** There are 4 profiles in complain mode and 91 profiles in unconfined mode.
 - `apparmor.aa_status.raw_output: 4 profiles are in complain mode., 91 profiles are in unconfined mode.`
- **Recommendation:** Review and set all AppArmor profiles to enforcing mode. For profiles in complain mode, analyze logs to identify and resolve any issues before switching to enforcing. For unconfined profiles, determine if they are necessary and if a profile should be created or loaded for them. Use `aa-enforce <profile_name>` to enforce profiles.

Rule 1.3.1.4: Ensure `apparmor_restrict_unprivileged_unconfined` is enabled

- **Status:** FAIL
- **Evidence:** The `kernel.apparmor_restrict_unprivileged_unconfined` sysctl parameter is not set to 1 at runtime or persisted.
 - `apparmor.restrict_unprivileged_sysctl.runtime_value: 0`
 - `apparmor.restrict_unprivileged_sysctl.persisted_value: null`
- **Recommendation:** Set `kernel.apparmor_restrict_unprivileged_unconfined` to 1.
 - Add or modify the line `kernel.apparmor_restrict_unprivileged_unconfined = 1` in a sysctl configuration file (e.g., `/etc/sysctl.d/99-apparmor.conf`).
 - Run `sudo sysctl -p /etc/sysctl.d/99-apparmor.conf` or `sudo sysctl -w kernel.apparmor_restrict_unprivileged_unconfined=1`

to apply the setting immediately.

Section 1.4 – Configure Bootloader

Rule 1.4.1: Ensure bootloader password is set

- **Status:** FAIL
- **Evidence:** No superusers or password lines were found in the GRUB configuration to protect the bootloader.

- ☐ `bootloader.password_config.has_superusers: false`
- ☐ `bootloader.password_config.has_password: false`

- **Recommendation:** Set a password for the GRUB bootloader.

1. Generate a GRUB password hash: `sudo grub-mkpasswd-pbkdf2`
2. Edit `/etc/grub.d/40_custom` and add the following lines, replacing `GRUB_PASSWORD` with your generated hash:

```
set superusers="root"  
password_pbkdf2 root GRUB_PASSWORD
```

3. Run `sudo update-grub` to apply changes.

Rule 1.4.2: Ensure access to bootloader config is configured

- **Status:** PASS
- **Evidence:** The `/boot/grub/grub.cfg` file has appropriate permissions (mode 0o600, owner root, group root).

- ☐ `bootloader.grub_cfg_permissions[0].mode_octal: 0o600`
- ☐ `bootloader.grub_cfg_permissions[0].owner: root`
- ☐ `bootloader.grub_cfg_permissions[0].group: root`

- **Recommendation:** None.

Section 1.5 – Configure Additional Process Hardening

Rule 1.5.1: Ensure fs.protected_hardlinks is configured

- **Status:** FAIL
- **Evidence:** The `fs.protected_hardlinks` sysctl parameter is set to 1 at runtime but is not persisted in a configuration file.

- ☐ `process_hardening.sysctl_parameters.fs.protected_hardlinks.runtime: 1`
- ☐ `process_hardening.sysctl_parameters.fs.protected_hardlinks.persisted: null`

- **Recommendation:** Add or modify the line `fs.protected_hardlinks = 1` in a sysctl configuration file (e.g., `/etc/sysctl.d/99-hardening.conf`). Run `sudo sysctl -p` to apply.

Rule 1.5.2: Ensure fs.protected_symlinks is configured

- **Status:** FAIL
- **Evidence:** The `fs.protected_symlinks` sysctl parameter is set to 1 at runtime but is not persisted in a configuration file.
 - `process_hardening.sysctl_parameters.fs.protected_symlinks.runtime_value` 1
 - `process_hardening.sysctl_parameters.fs.protected_symlinks.persisted_value` null
- **Recommendation:** Add or modify the line `fs.protected_symlinks = 1` in a sysctl configuration file (e.g., `/etc/sysctl.d/99-hardening.conf`). Run `sudo sysctl -p` to apply.

Rule 1.5.3: Ensure kernel.yama.ptrace_scope is configured

- **Status:** PASS
- **Evidence:** The `kernel.yama.ptrace_scope` sysctl parameter is set to 1 at runtime and is persisted in `/etc/sysctl.d/10-ptrace.conf`.
 - `process_hardening.sysctl_parameters.kernel.yama.ptrace_scope.runtime_value` 1
 - `process_hardening.sysctl_parameters.kernel.yama.ptrace_scope.persisted_value` 1
 - `process_hardening.sysctl_parameters.kernel.yama.ptrace_scope.persisted_file` `/etc/sysctl.d/10-ptrace.conf`
- **Recommendation:** None.

Rule 1.5.4: Ensure fs.suid_dumpable is configured

- **Status:** FAIL
- **Evidence:** The `fs.suid_dumpable` sysctl parameter is set to 2 at runtime, but the recommended value is 0. It is also not persisted.
 - `process_hardening.sysctl_parameters.fs.suid_dumpable.runtime_value` 2
 - `process_hardening.sysctl_parameters.fs.suid_dumpable.persisted_value` null
- **Recommendation:** Set `fs.suid_dumpable` to 0.
 - Add or modify the line `fs.suid_dumpable = 0` in a sysctl configuration file (e.g., `/etc/sysctl.d/99-hardening.conf`).
 - Run `sudo sysctl -p` to apply.

Rule 1.5.5: Ensure kernel.dmesg_restrict is configured

- **Status:** FAIL
- **Evidence:** The `kernel.dmesg_restrict` sysctl parameter is set to 1 at runtime but is not persisted in a configuration file.
 - `process_hardening.sysctl_parameters.kernel.dmesg_restrict.runtime_value` 1

- 1
- `process_hardening.sysctl_parameters.kernel.dmesg_restrict.persist`
`null`

- **Recommendation:** Add or modify the line `kernel.dmesg_restrict = 1` in a `sysctl` configuration file (e.g., `/etc/sysctl.d/99-hardening.conf`). Run `sudo sysctl -p` to apply.

Rule 1.5.6: Ensure prelink is not installed

- **Status:** PASS
- **Evidence:** The `prelink` package is not installed on the system.
 - `process_hardening.prelink.prelink_installed: false`
- **Recommendation:** None.

Rule 1.5.7: Ensure Automatic Error Reporting is configured

- **Status:** FAIL
- **Evidence:** The `apport` package is installed, and the service is enabled and active. The `/etc/default/apport` setting is `enabled=1`.
 - `process_hardening.apport.apport_installed: true`
 - `process_hardening.apport.service_enabled: enabled`
 - `process_hardening.apport.service_active: active`
 - `process_hardening.apport.etc_default_setting: enabled=1`
- **Recommendation:** Disable automatic error reporting.
 1. Edit `/etc/default/apport` and set `enabled=0`.
 2. Run `sudo systemctl disable apport.service` and `sudo systemctl stop apport.service`.

Rule 1.5.8: Ensure kernel.kptr_restrict is configured

- **Status:** PASS
- **Evidence:** The `kernel.kptr_restrict` `sysctl` parameter is set to 1 at runtime and is persisted in `/etc/sysctl.d/10-kernel-hardening.conf`. CIS recommends 2 for Level 2, but 1 is acceptable for Level 1.
 - `process_hardening.sysctl_parameters.kernel.kptr_restrict.runtime`
`1`
 - `process_hardening.sysctl_parameters.kernel.kptr_restrict.persist`
`1`
 - `process_hardening.sysctl_parameters.kernel.kptr_restrict.persist`
`/etc/sysctl.d/10-kernel-hardening.conf`
- **Recommendation:** None (for Level 1). For Level 2, consider setting to 2 by adding `kernel.kptr_restrict = 2` to a `sysctl` configuration file and running `sudo sysctl -p`.

Rule 1.5.9: Ensure kernel.randomize_va_space is configured

- **Status:** FAIL
- **Evidence:** The `kernel.randomize_va_space` `sysctl` parameter is set to 2 at runtime but is not persisted in a configuration file.
 - `process_hardening.sysctl_parameters.kernel.randomize_va_space.r`
2
 - `process_hardening.sysctl_parameters.kernel.randomize_va_space.r`
null
- **Recommendation:** Add or modify the line `kernel.randomize_va_space = 2` in a `sysctl` configuration file (e.g., `/etc/sysctl.d/99-hardening.conf`). Run `sudo sysctl -p` to apply.

Rule 1.5.10: Ensure core file size is configured

- **Status:** FAIL
- **Evidence:** The `ulimit -Sc` command shows a soft core file size limit of 0, but no `hard core 0` entry was found in `/etc/security/limits.conf`.
 - `process_hardening.core_dumps.ulimit_soft_core:0`
 - `process_hardening.core_dumps.limits_conf_lines: []`
- **Recommendation:** Add the following line to `/etc/security/limits.conf` to ensure core dumps are disabled for all users:


```
* hard core 0
```

Rule 1.5.11: Ensure systemd-coredump ProcessSizeMax is configured

- **Status:** FAIL
- **Evidence:** No `ProcessSizeMax=0` entry was found in `systemd-coredump` configuration.
 - `process_hardening.core_dumps.systemd_coredump_process_size_max:`
[]
- **Recommendation:** Create or modify `/etc/systemd/coredump.conf.d/99-disable-coredumps.conf` with the following content:


```
[Coredump]
ProcessSizeMax=0
```

Then run `sudo systemctl daemon-reload`.

Rule 1.5.12: Ensure systemd-coredump Storage is configured

- **Status:** FAIL
- **Evidence:** No `Storage=none` entry was found in `systemd-coredump` configuration.
 - `process_hardening.core_dumps.systemd_coredump_storage: []`
- **Recommendation:** Create or modify `/etc/systemd/coredump.conf.d/99-disable-coredumps.conf` with the following content:

```
[Coredump]  
Storage=none
```

Then run `sudo systemctl daemon-reload`.

Section 1.6 – Configure Command Line Warning Banners

Rule 1.6.1: Ensure `/etc/motd` is configured

- **Status:** FAIL
- **Evidence:** The `/etc/motd` file does not exist. Dynamic MOTD scripts are in use, which may display system information.
 - `warning_banners.motd.exists: false`
 - `warning_banners.pam_motd.update_motd_files_content: Contains scripts like 00-header, 50-landscape-sysinfo, 90-updates-available which display system information.`
- **Recommendation:** Create a compliant `/etc/motd` file with a warning banner. Ensure that dynamic MOTD scripts do not display sensitive system information. If dynamic MOTD is desired, ensure it only displays approved information.

Rule 1.6.2: Ensure `/etc/issue` is configured

- **Status:** FAIL
- **Evidence:** The `/etc/issue` file contains OS and kernel release information (Ubuntu 24.04.2 LTS \n \l). It should contain only an authorized warning banner.
 - `warning_banners.issue.content: Ubuntu 24.04.2 LTS \n \l\n\n`
- **Recommendation:** Edit `/etc/issue` to display only an authorized warning banner, without any system information or escape sequences. For example:

```
Authorized uses only. All activity may be monitored and reported.
```

Rule 1.6.3: Ensure `/etc/issue.net` is configured

- **Status:** FAIL
- **Evidence:** The `/etc/issue.net` file contains OS release information (Ubuntu 24.04.2 LTS). It should contain only an authorized warning banner.

- `warning_banners.issue_net.content: Ubuntu 24.04.2 LTS\n`

- **Recommendation:** Edit `/etc/issue.net` to display only an authorized warning banner, without any system information or escape sequences. For example:

```
Authorized uses only. All activity may be monitored and reported.
```

Rule 1.6.4: Ensure `pam_motd` is configured

- **Status:** FAIL
- **Evidence:** The `pam_motd.so` module is configured in `/etc/pam.d/login` and /

`etc/pam.d/sshd` to display dynamic MOTD content from `/run/motd.dynamic`. This dynamic content is generated by scripts in `/etc/update-motd.d/` which include system information.

- `warning_banners.pam_motd.pam_references`: Shows `pam_motd.so` in login and `sshd` PAM files.
- `warning_banners.pam_motd.update_motd_files_content`: Contains scripts that generate system information.
- **Recommendation:** Disable or modify the dynamic MOTD generation scripts in `/etc/update-motd.d/` to ensure no sensitive system information is displayed. Alternatively, remove the `pam_motd.so` lines from `/etc/pam.d/login` and `/etc/pam.d/sshd` if a static banner is preferred.

Rule 1.6.6: Ensure access to `/etc/motd` is configured

- **Status:** FAIL
- **Evidence:** The `/etc/motd` file does not exist.
 - `warning_banners.motd.exists`: false
- **Recommendation:** Create the `/etc/motd` file and ensure its permissions are `0o644` or more restrictive, owned by `root:root`.

Rule 1.6.7: Ensure access to `/etc/issue` is configured

- **Status:** PASS
- **Evidence:** The `/etc/issue` file has appropriate permissions (mode `0o644`, owner `root`, group `root`).
 - `warning_banners.issue.stat.mode_octal`: `0o644`
 - `warning_banners.issue.stat.owner`: `root`
 - `warning_banners.issue.stat.group`: `root`
- **Recommendation:** None.

Rule 1.6.8: Ensure access to `/etc/issue.net` is configured

- **Status:** PASS
- **Evidence:** The `/etc/issue.net` file has appropriate permissions (mode `0o644`, owner `root`, group `root`).
 - `warning_banners.issue_net.stat.mode_octal`: `0o644`
 - `warning_banners.issue_net.stat.owner`: `root`
 - `warning_banners.issue_net.stat.group`: `root`
- **Recommendation:** None.

Rule 1.6.9: Ensure access to `pam_motd` file is configured

- **Status:** PASS
- **Evidence:** All files in `/etc/update-motd.d/` have appropriate permissions (mode `0o755` or `0o644`, owner `root`, group `root`).

○ Example: /etc/update-motd.d/00-header has 0o755 root root.

- **Recommendation:** None.

Section 1.7 – Configure GNOME Display Manager

Rule 1.7.1: Ensure GDM login banner is configured

- **Status:** NOT APPLICABLE
- **Evidence:** GDM (Gnome Display Manager) is not installed on the system.

○ `gnome.gdm_installed: false`

- **Recommendation:** None.

Rule 1.7.2: Ensure GDM disable-user-list is configured

- **Status:** NOT APPLICABLE
- **Evidence:** GDM (Gnome Display Manager) is not installed on the system.

○ `gnome.gdm_installed: false`

- **Recommendation:** None.

Rule 1.7.3: Ensure GDM screen lock is configured

- **Status:** NOT APPLICABLE
- **Evidence:** GDM (Gnome Display Manager) is not installed on the system.

○ `gnome.gdm_installed: false`

- **Recommendation:** None.

Rule 1.7.4: Ensure GDM automount is configured

- **Status:** NOT APPLICABLE
- **Evidence:** GDM (Gnome Display Manager) is not installed on the system.

○ `gnome.gdm_installed: false`

- **Recommendation:** None.

Rule 1.7.5: Ensure GDM autorun-never is configured

- **Status:** NOT APPLICABLE
- **Evidence:** GDM (Gnome Display Manager) is not installed on the system.

○ `gnome.gdm_installed: false`

- **Recommendation:** None.

Rule 1.7.6: Ensure XDMCP is not enabled

- **Status:** NOT APPLICABLE
- **Evidence:** GDM (Gnome Display Manager) is not installed on the system.

☐ `gnome.gdm_installed: false`

- **Recommendation:** None.

Rule 1.7.7: Ensure Xwayland is configured

- **Status:** NOT APPLICABLE
- **Evidence:** GDM (Gnome Display Manager) is not installed on the system.

☐ `gnome.gdm_installed: false`

- **Recommendation:** None.

Section 2 – Services

Rule 2.1.1: Ensure autofs services are not in use

- **Status:** PASS
- **Evidence:** The autofs package is not installed and no autofs units are active.

☐ `services.server_services[0].any_package_installed: false`

☐ `services.server_services[0].any_unit_active: false`

- **Recommendation:** None.

Rule 2.1.2: Ensure mail transfer agents are configured for local-only mode

- **Status:** PASS
- **Evidence:** No Mail Transfer Agent (MTA) packages (postfix, exim4, sendmail) are installed on the system.

☐ `services.server_services[1].any_package_installed: false`

☐ `services.server_services[1].mta_detected: none`

- **Recommendation:** None.

Rule 2.1.3: Ensure avahi daemon services are not in use

- **Status:** PASS
- **Evidence:** The avahi-daemon package is not installed and no avahi-daemon units are active.

☐ `services.server_services[2].any_package_installed: false`

☐ `services.server_services[2].any_unit_active: false`

- **Recommendation:** None.

Rule 2.1.4: Ensure only approved services are listening on a network interface

- **Status:** INFORMATIONAL
- **Evidence:** The following services are listening on network interfaces:
 - `udp 127.0.0.54:53 (systemd-resolve)`
 - `udp 127.0.0.53%lo:53 (systemd-resolve)`
 - `udp 10.100.0.22%enp0s3:68 (systemd-network)`
 - `tcp 0.0.0.0:443 (node)`
 - `tcp 0.0.0.0:1514 (wazuh-remoted)`
 - `tcp 0.0.0.0:1515 (wazuh-authd)`
 - `tcp 0.0.0.0:22 (sshd)`
 - `tcp 0.0.0.0:23 (inetd)`
 - `tcp 127.0.0.53%lo:53 (systemd-resolve)`
 - `tcp 127.0.0.54:53 (systemd-resolve)`
 - `tcp 0.0.0.0:55000 (python3)`
 - `tcp *:21 (vsftpd)`
 - `tcp [::]:22 (sshd)`
 - `tcp [::]:23 (inetd)`
 - `tcp [::ffff:127.0.0.1]:9300 (java)`
 - `tcp [::ffff:127.0.0.1]:9200 (java)`
- **Recommendation:** Review the list of listening services and ensure that each service is explicitly authorized and necessary for the system's function. Specifically, `vsftpd` (FTP server) and `inetd` (Telnet server) are active and listening on public interfaces, which are generally not recommended for security. The `node` process listening on port 443 and `python3` on 55000 should also be verified.

Rule 2.1.5: Ensure dhcp server services are not in use

- **Status:** PASS
- **Evidence:** The `isc-dhcp-server` package is not installed and no DHCP server units are active.
 - `services.server_services[4].any_package_installed: false`
 - `services.server_services[4].any_unit_active: false`
- **Recommendation:** None.

Rule 2.1.6: Ensure web server services are not in use

- **Status:** FAIL
- **Evidence:** No common web server packages (`apache2`, `nginx`, `lighttpd`) are installed, and no associated units are active. However, a `node` process is listening on port 443, indicating an active web server.
 - `services.server_services[5].any_package_installed: false`
 - `services.server_services[5].any_unit_active: false`
 - `services.listening_sockets_raw: tcp LISTEN 0 511 0.0.0.0:443 0.0.0.0:* users:(("node",pid=698,fd=18))`
- **Recommendation:** Identify the `node` process listening on port 443. If it is an unauthorized web server, disable or remove it. If it is an authorized web server, ensure it is properly secured and configured according to best practices.

Rule 2.1.7: Ensure dns server services are not in use

- **Status:** PASS
- **Evidence:** The bind9 package is not installed and no DNS server units are active. systemd-resolve is listening on local interfaces for DNS, which is expected for a client.
 - services.server_services[6].any_package_installed: false
 - services.server_services[6].any_unit_active: false
- **Recommendation:** None.

Rule 2.1.8: Ensure ftp server services are not in use

- **Status:** FAIL
- **Evidence:** The vsftpd package is installed, enabled, and active, and is listening on port 21.
 - services.server_services[7].any_package_installed: true
 - services.server_services[7].units_status[0].enabled: enabled
 - services.server_services[7].units_status[0].active: active
 - services.listening_sockets_raw: tcp LISTEN 0 32 *:21 *:*
 - users: (("vsftpd", pid=706, fd=3))
- **Recommendation:** Unless explicitly required, FTP services should be disabled and uninstalled due to their insecure nature (transmitting credentials in plaintext).
 1. Disable the vsftpd service: `sudo systemctl disable vsftpd.service`
 2. Stop the vsftpd service: `sudo systemctl stop vsftpd.service`
 3. Uninstall the package: `sudo apt purge vsftpd`

Rule 2.1.9: Ensure dnsmasq services are not in use

- **Status:** PASS
- **Evidence:** The dnsmasq package is not installed and no dnsmasq units are active.
 - services.server_services[8].any_package_installed: false
 - services.server_services[8].any_unit_active: false
- **Recommendation:** None.

Rule 2.1.10: Ensure ldap server services are not in use

- **Status:** PASS
- **Evidence:** The slapd package is not installed and no slapd units are active.
 - services.server_services[9].any_package_installed: false
 - services.server_services[9].any_unit_active: false
- **Recommendation:** None.

Rule 2.1.11: Ensure message access server services are not in use

- **Status:** PASS

- **Evidence:** No message access server packages (dovecot-imapd, dovecot-pop3d, cyrus-imapd) are installed and no dovecot units are active.
 - `services.server_services[10].any_package_installed: false`
 - `services.server_services[10].any_unit_active: false`
- **Recommendation:** None.

Rule 2.1.12: Ensure network file system services are not in use

- **Status:** PASS
- **Evidence:** The `nfs-kernel-server` package is not installed and no `nfs-kernel-server` units are active.
 - `services.server_services[11].any_package_installed: false`
 - `services.server_services[11].any_unit_active: false`
- **Recommendation:** None.

Rule 2.1.13: Ensure nis server services are not in use

- **Status:** PASS
- **Evidence:** The `nis` package is not installed and no `ypserv` units are active.
 - `services.server_services[12].any_package_installed: false`
 - `services.server_services[12].any_unit_active: false`
- **Recommendation:** None.

Rule 2.1.14: Ensure print server services are not in use

- **Status:** PASS
- **Evidence:** The `cups` package is not installed and no `cups` units are active.
 - `services.server_services[13].any_package_installed: false`
 - `services.server_services[13].any_unit_active: false`
- **Recommendation:** None.

Rule 2.1.15: Ensure rpcbind services are not in use

- **Status:** PASS
- **Evidence:** The `rpcbind` package is not installed and no `rpcbind` units are active.
 - `services.server_services[14].any_package_installed: false`
 - `services.server_services[14].any_unit_active: false`
- **Recommendation:** None.

Rule 2.1.16: Ensure rsync services are not in use

- **Status:** PASS
- **Evidence:** The `rsync` package is installed, but the `rsync.service` unit is disabled and inactive.

- `services.server_services[15].any_package_installed: true`
- `services.server_services[15].units_status[0].enabled: disabled`
- `services.server_services[15].units_status[0].active: inactive`

- **Recommendation:** None.

Rule 2.1.17: Ensure samba file server services are not in use

- **Status:** PASS
- **Evidence:** The samba package is not installed and no smbd units are active.
 - `services.server_services[16].any_package_installed: false`
 - `services.server_services[16].any_unit_active: false`

- **Recommendation:** None.

Rule 2.1.18: Ensure snmp services are not in use

- **Status:** PASS
- **Evidence:** The snmpd package is not installed and no snmpd units are active.
 - `services.server_services[17].any_package_installed: false`
 - `services.server_services[17].any_unit_active: false`

- **Recommendation:** None.

Rule 2.1.19: Ensure telnet server services are not in use

- **Status:** FAIL
- **Evidence:** The telnetd and inetutils-telnetd packages are installed. The inetd.service and openbsd-inetd.service units are enabled and active, and listening on port 23.
 - `services.server_services[18].any_package_installed: true`
 - `services.server_services[18].units_status[2].enabled: enabled`
 - `services.server_services[18].units_status[2].active: active`
 - `services.listening_sockets_raw: tcp LISTEN 0 128 0.0.0.0:23 0.0.0.0:* users:(("inetd",pid=760,fd=7))`
- **Recommendation:** Telnet services transmit data, including credentials, in plaintext and should be disabled and uninstalled.

1. Disable the inetd.service: `sudo systemctl disable inetd.service`
2. Stop the inetd.service: `sudo systemctl stop inetd.service`
3. Uninstall the packages: `sudo apt purge telnetd inetutils-telnetd`

Rule 2.1.20: Ensure tftp server services are not in use

- **Status:** PASS
- **Evidence:** No TFTP server packages (tftpd, tftpd-hpa, atftpd) are installed and no TFTP server units are active.
 - `services.server_services[19].any_package_installed: false`

○ `services.server_services[19].any_unit_active: false`

- **Recommendation:** None.

Rule 2.1.21: Ensure web proxy server services are not in use

- **Status:** PASS
- **Evidence:** No web proxy server packages (squid, squid3) are installed and no squid units are active.

○ `services.server_services[20].any_package_installed: false`

○ `services.server_services[20].any_unit_active: false`

- **Recommendation:** None.

Rule 2.1.22: Ensure xinetd services are not in use

- **Status:** PASS
- **Evidence:** The xinetd package is not installed and no xinetd units are active.

○ `services.server_services[21].any_package_installed: false`

○ `services.server_services[21].any_unit_active: false`

- **Recommendation:** None.

Rule 2.1.23: Ensure X window server services are not in use

- **Status:** PASS
- **Evidence:** No X window server packages (xserver-xorg, xserver-xorg-core, xserver-common) are installed, and no Xorg processes or display manager units are active.

○ `services.server_services[22].any_package_installed: false`

○ `services.server_services[22].xorg_process_running: false`

○ `services.server_services[22].units_status: All inactive/not-found.`

- **Recommendation:** None.

Rule 2.2.1: Ensure nis client is not installed

- **Status:** PASS
- **Evidence:** The nis client package is not installed.

○ `services.client_services[0].any_package_installed: false`

- **Recommendation:** None.

Rule 2.2.2: Ensure rsh client is not installed

- **Status:** PASS
- **Evidence:** The rsh-client package is not installed.

○ `services.client_services[1].any_package_installed: false`

- **Recommendation:** None.

Rule 2.2.3: Ensure talk client is not installed

- **Status:** PASS
- **Evidence:** The `talk` client package is not installed.

○ `services.client_services[2].any_package_installed: false`

- **Recommendation:** None.

Rule 2.2.4: Ensure telnet client is not installed

- **Status:** FAIL
- **Evidence:** The `telnet` and `inetutils-telnet` client packages are installed.

○ `services.client_services[3].any_package_installed: true`

- **Recommendation:** Uninstall the `telnet` client packages: `sudo apt purge telnet inetutils-telnet`.

Rule 2.2.5: Ensure ldap client is not installed

- **Status:** PASS
- **Evidence:** The `ldap-utils` client package is not installed.

○ `services.client_services[4].any_package_installed: false`

- **Recommendation:** None.

Rule 2.2.6: Ensure ftp client is not installed

- **Status:** FAIL
- **Evidence:** The `ftp` client package is installed.

○ `services.client_services[5].any_package_installed: true`

- **Recommendation:** Uninstall the `ftp` client package: `sudo apt purge ftp`.

Rule 2.3.1.1: Ensure a single time synchronization daemon is in use

- **Status:** PASS
- **Evidence:** Only one time synchronization daemon, `systemd-timesyncd`, is active.

○ `time_sync.time_sync_general.active_daemons: ['systemd-timesyncd']`

○ `time_sync.time_sync_general.active_daemon_count: 1`

- **Recommendation:** None.

Rule 2.3.2.1: Ensure systemd-timesyncd configured with authorized timeserver

- **Status:** FAIL
- **Evidence:** `systemd-timesyncd` is active, but no NTP servers are explicitly

configured in its configuration file.

- `time_sync.systemd-timesyncd.service.active: active`
- `time_sync.systemd-timesyncd.has_ntp_configured: false`

- **Recommendation:** Configure `systemd-timesyncd` to use an authorized NTP server.

1. Edit `/etc/systemd/timesyncd.conf` and add or modify the `NTP=` line with your authorized NTP servers (e.g., `NTP=ntp.example.com`).
2. Restart the service: `sudo systemctl restart systemd-timesyncd.service`.

Rule 2.3.2.2: Ensure `systemd-timesyncd` is enabled and running

- **Status:** PASS
- **Evidence:** The `systemd-timesyncd.service` is enabled and active.

- `time_sync.systemd-timesyncd.service.enabled: enabled`
- `time_sync.systemd-timesyncd.service.active: active`

- **Recommendation:** None.

Rule 2.3.3.1: Ensure `chrony` is configured

- **Status:** NOT APPLICABLE
- **Evidence:** `chrony` is not installed or active on the system. `systemd-timesyncd` is the active time synchronization daemon.

- `time_sync.chrony.service.active: inactive`

- **Recommendation:** None.

Rule 2.3.3.2: Ensure `chrony` is running as user `_chrony`

- **Status:** NOT APPLICABLE
- **Evidence:** `chrony` is not installed or active on the system.

- `time_sync.chrony.service.active: inactive`

- **Recommendation:** None.

Rule 2.3.3.3: Ensure `chrony` is enabled and running

- **Status:** NOT APPLICABLE
- **Evidence:** `chrony` is not installed or active on the system.

- `time_sync.chrony.service.active: inactive`

- **Recommendation:** None.

Rule 2.4.1.1: Ensure `cron` daemon is enabled and active

- **Status:** PASS
- **Evidence:** The `cron` package is installed, and the `cron.service` is enabled and

active.

- `job_schedulers.cron.cron_installed: true`
- `job_schedulers.cron.cron_service.enabled: enabled`
- `job_schedulers.cron.cron_service.active: active`

- **Recommendation:** None.

Rule 2.4.1.2: Ensure access to /etc/crontab is configured

- **Status:** PASS
- **Evidence:** The `/etc/crontab` file has appropriate permissions (mode 0o644, owner root, group root).

- `job_schedulers.cron.cron_paths_permissions[0].mode_octal: 0o644`
- `job_schedulers.cron.cron_paths_permissions[0].owner: root`
- `job_schedulers.cron.cron_paths_permissions[0].group: root`

- **Recommendation:** None.

Rule 2.4.1.3: Ensure access to /etc/cron.hourly is configured

- **Status:** PASS
- **Evidence:** The `/etc/cron.hourly` directory has appropriate permissions (mode 0o755, owner root, group root).

- `job_schedulers.cron.cron_paths_permissions[1].mode_octal: 0o755`
- `job_schedulers.cron.cron_paths_permissions[1].owner: root`
- `job_schedulers.cron.cron_paths_permissions[1].group: root`

- **Recommendation:** None.

Rule 2.4.1.4: Ensure access to /etc/cron.daily is configured

- **Status:** PASS
- **Evidence:** The `/etc/cron.daily` directory has appropriate permissions (mode 0o755, owner root, group root).

- `job_schedulers.cron.cron_paths_permissions[2].mode_octal: 0o755`
- `job_schedulers.cron.cron_paths_permissions[2].owner: root`
- `job_schedulers.cron.cron_paths_permissions[2].group: root`

- **Recommendation:** None.

Rule 2.4.1.5: Ensure access to /etc/cron.weekly is configured

- **Status:** PASS
- **Evidence:** The `/etc/cron.weekly` directory has appropriate permissions (mode 0o755, owner root, group root).

- `job_schedulers.cron.cron_paths_permissions[3].mode_octal:`

- 0o755
- job_schedulers.cron.cron_paths_permissions[3].owner: root
- job_schedulers.cron.cron_paths_permissions[3].group: root

- **Recommendation:** None.

Rule 2.4.1.6: Ensure access to /etc/cron.monthly is configured

- **Status:** PASS
- **Evidence:** The /etc/cron.monthly directory has appropriate permissions (mode 0o755, owner root, group root).

- job_schedulers.cron.cron_paths_permissions[4].mode_octal: 0o755
- job_schedulers.cron.cron_paths_permissions[4].owner: root
- job_schedulers.cron.cron_paths_permissions[4].group: root

- **Recommendation:** None.

Rule 2.4.1.7: Ensure access to /etc/cron.yearly is configured

- **Status:** PASS
- **Evidence:** The /etc/cron.yearly directory has appropriate permissions (mode 0o755, owner root, group root).

- job_schedulers.cron.cron_paths_permissions[5].mode_octal: 0o755
- job_schedulers.cron.cron_paths_permissions[5].owner: root
- job_schedulers.cron.cron_paths_permissions[5].group: root

- **Recommendation:** None.

Rule 2.4.1.8: Ensure access to /etc/cron.d is configured

- **Status:** PASS
- **Evidence:** The /etc/cron.d directory has appropriate permissions (mode 0o755, owner root, group root).

- job_schedulers.cron.cron_paths_permissions[6].mode_octal: 0o755
- job_schedulers.cron.cron_paths_permissions[6].owner: root
- job_schedulers.cron.cron_paths_permissions[6].group: root

- **Recommendation:** None.

Rule 2.4.1.9: Ensure access to crontab is configured

- **Status:** PASS
- **Evidence:** The /var/spool/cron/crontabs directory has mode 0o1730, owner root, group crontab. The /usr/bin/crontab binary has mode 0o2755, owner root, group crontab. These permissions are compliant.

- job_schedulers.cron.crontabs_spool_stat.mode_octal: 0o1730
- job_schedulers.cron.crontabs_spool_stat.owner: root

- `job_schedulers.cron.crontabs_spool_stat.group: crontab`
- `job_schedulers.cron.crontab_binary_stat.mode_octal: 0o2755`
- `job_schedulers.cron.crontab_binary_stat.owner: root`
- `job_schedulers.cron.crontab_binary_stat.group: crontab`

- **Recommendation:** None.

Rule 2.4.2.1: Ensure access to at is configured

- **Status:** NOT APPLICABLE
- **Evidence:** The at package is not installed on the system.

- `job_schedulers.at.at_installed: false`

- **Recommendation:** None.

Section 3 – Network Configuration

Rule 3.1.1: Ensure IPv6 status is identified

- **Status:** INFORMATIONAL
- **Evidence:** IPv6 is enabled on the system.

- `network_config.network_devices.ipv6_status.sysfs_disable_value: 0 (IPv6 enabled)`
- `network_config.network_devices.ipv6_status.grub_ipv6_disable_flag: false (No ipv6.disable=1 in GRUB)`
- `network_config.network_devices.ipv6_status.inet6_address_count: 2 (IPv6 addresses found on interfaces)`

- **Recommendation:** None.

Rule 3.1.2: Ensure wireless interfaces are not available

- **Status:** FAIL
- **Evidence:** A wireless kernel module (`cfg80211`) is loaded, indicating wireless capabilities are available.

- `network_config.network_devices.wireless_interfaces.loaded_wireless_interfaces: ['cfg80211']`
- `network_config.network_devices.wireless_interfaces.any_wireless_interfaces_loaded: true`

- **Recommendation:** If wireless functionality is not required, blacklist the `cfg80211` module (and any other wireless-related modules) to prevent its loading.

1. Create or edit `/etc/modprobe.d/blacklist-wireless.conf` and add:

```
blacklist cfg80211
```

2. Run `sudo update-initramfs -u` and reboot.

Rule 3.1.3: Ensure bluetooth services are not in use

- **Status:** PASS
- **Evidence:** The `bluez` package is not installed, and the Bluetooth service is inactive.
 - `network_config.network_devices.bluetooth.bluez_installed:`
`false`
 - `network_config.network_devices.bluetooth.bluetooth_service_acti`
`inactive`
- **Recommendation:** None.

Rule 3.2.1: Ensure atm kernel module is not available

- **Status:** PASS
- **Evidence:** The `atm` module is not currently loaded and no modprobe configuration lines were found.
 - `network_config.kernel_modules.atm.currently_loaded:` `false`
 - `network_config.kernel_modules.atm.modprobe_config_lines:` `[]`
- **Recommendation:** None.

Rule 3.2.2: Ensure can kernel module is not available

- **Status:** PASS
- **Evidence:** The `can` module is not currently loaded and no modprobe configuration lines were found.
 - `network_config.kernel_modules.can.currently_loaded:` `false`
 - `network_config.kernel_modules.can.modprobe_config_lines:` `[]`
- **Recommendation:** None.

Rule 3.2.3: Ensure dccp kernel module is not available

- **Status:** PASS
- **Evidence:** The `dccp` module is not currently loaded and no modprobe configuration lines were found.
 - `network_config.kernel_modules.dccp.currently_loaded:` `false`
 - `network_config.kernel_modules.dccp.modprobe_config_lines:` `[]`
- **Recommendation:** None.

Rule 3.2.4: Ensure rds kernel module is not available

- **Status:** PASS
- **Evidence:** The `rds` module is not currently loaded and no modprobe configuration lines were found.
 - `network_config.kernel_modules.rds.currently_loaded:` `false`
 - `network_config.kernel_modules.rds.modprobe_config_lines:` `[]`
- **Recommendation:** None.

Rule 3.2.5: Ensure sctp kernel module is not available

- **Status:** PASS
- **Evidence:** The `sctp` module is not currently loaded and no modprobe configuration lines were found.
 - `network_config.kernel_modules.sctp.currently_loaded: false`
 - `network_config.kernel_modules.sctp.modprobe_config_lines: []`
- **Recommendation:** None.

Rule 3.2.6: Ensure tipc kernel module is not available

- **Status:** PASS
- **Evidence:** The `tipc` module is not currently loaded and no modprobe configuration lines were found.
 - `network_config.kernel_modules.tipc.currently_loaded: false`
 - `network_config.kernel_modules.tipc.modprobe_config_lines: []`
- **Recommendation:** None.

Rule 3.3.1.1: Ensure net.ipv4.ip_forward is configured

- **Status:** PASS
- **Evidence:** The `net.ipv4.ip_forward` parameter is set to 0 at runtime and is not explicitly persisted (default is 0).
 - `network_config.sysctl_ipv4.net.ipv4.ip_forward.runtime_value: 0`
 - `network_config.sysctl_ipv4.net.ipv4.ip_forward.persisted_value: null`
- **Recommendation:** None.

Rule 3.3.1.2: Ensure net.ipv4.conf.all.forwarding is configured

- **Status:** PASS
- **Evidence:** The `net.ipv4.conf.all.forwarding` parameter is set to 0 at runtime and is not explicitly persisted (default is 0).
 - `network_config.sysctl_ipv4.net.ipv4.conf.all.forwarding.runtime_value: 0`
 - `network_config.sysctl_ipv4.net.ipv4.conf.all.forwarding.persisted_value: null`
- **Recommendation:** None.

Rule 3.3.1.3: Ensure net.ipv4.conf.default.forwarding is configured

- **Status:** PASS
- **Evidence:** The `net.ipv4.conf.default.forwarding` parameter is set to 0 at runtime and is not explicitly persisted (default is 0).

- `network_config.sysctl_ipv4.net.ipv4.conf.default.forwarding.rur`
0
- `network_config.sysctl_ipv4.net.ipv4.conf.default.forwarding.per`
null

- **Recommendation:** None.

Rule 3.3.1.4: Ensure net.ipv4.conf.all.send_redirects is configured

- **Status:** FAIL
- **Evidence:** The `net.ipv4.conf.all.send_redirects` parameter is set to 1 at runtime, but the recommended value is 0. It is also not persisted.

- `network_config.sysctl_ipv4.net.ipv4.conf.all.send_redirects.rur`
1
- `network_config.sysctl_ipv4.net.ipv4.conf.all.send_redirects.per`
null

- **Recommendation:** Set `net.ipv4.conf.all.send_redirects` to 0.

- Add or modify the line `net.ipv4.conf.all.send_redirects = 0` in a sysctl configuration file (e.g., `/etc/sysctl.d/99-network.conf`).
- Run `sudo sysctl -p` to apply.

Rule 3.3.1.5: Ensure net.ipv4.conf.default.send_redirects is configured

- **Status:** FAIL
- **Evidence:** The `net.ipv4.conf.default.send_redirects` parameter is set to 1 at runtime, but the recommended value is 0. It is also not persisted.

- `network_config.sysctl_ipv4.net.ipv4.conf.default.send_redirects`
1
- `network_config.sysctl_ipv4.net.ipv4.conf.default.send_redirects`
null

- **Recommendation:** Set `net.ipv4.conf.default.send_redirects` to 0.

- Add or modify the line `net.ipv4.conf.default.send_redirects = 0` in a sysctl configuration file (e.g., `/etc/sysctl.d/99-network.conf`).
- Run `sudo sysctl -p` to apply.

Rule 3.3.1.6: Ensure net.ipv4.icmp_ignore_bogus_error_responses is configured

- **Status:** PASS
- **Evidence:** The `net.ipv4.icmp_ignore_bogus_error_responses` parameter is set to 1 at runtime and is not explicitly persisted (default is 1).

- `network_config.sysctl_ipv4.net.ipv4.icmp_ignore_bogus_error_res`
1
- `network_config.sysctl_ipv4.net.ipv4.icmp_ignore_bogus_error_res`
null

- **Recommendation:** None.

Rule 3.3.1.7: Ensure `net.ipv4.icmp_echo_ignore_broadcasts` is configured

- **Status:** PASS
- **Evidence:** The `net.ipv4.icmp_echo_ignore_broadcasts` parameter is set to 1 at runtime and is not explicitly persisted (default is 1).
 - `network_config.sysctl_ipv4.net.ipv4.icmp_echo_ignore_broadcasts 1`
 - `network_config.sysctl_ipv4.net.ipv4.icmp_echo_ignore_broadcasts null`
- **Recommendation:** None.

Rule 3.3.1.8: Ensure `net.ipv4.conf.all.accept_redirects` is configured

- **Status:** FAIL
- **Evidence:** The `net.ipv4.conf.all.accept_redirects` parameter is set to 1 at runtime, but the recommended value is 0. It is also not persisted.
 - `network_config.sysctl_ipv4.net.ipv4.conf.all.accept_redirects.r 1`
 - `network_config.sysctl_ipv4.net.ipv4.conf.all.accept_redirects.p null`
- **Recommendation:** Set `net.ipv4.conf.all.accept_redirects` to 0.
 - Add or modify the line `net.ipv4.conf.all.accept_redirects = 0` in a sysctl configuration file (e.g., `/etc/sysctl.d/99-network.conf`).
 - Run `sudo sysctl -p` to apply.

Rule 3.3.1.9: Ensure `net.ipv4.conf.default.accept_redirects` is configured

- **Status:** FAIL
- **Evidence:** The `net.ipv4.conf.default.accept_redirects` parameter is set to 1 at runtime, but the recommended value is 0. It is also not persisted.
 - `network_config.sysctl_ipv4.net.ipv4.conf.default.accept_redirec 1`
 - `network_config.sysctl_ipv4.net.ipv4.conf.default.accept_redirec null`
- **Recommendation:** Set `net.ipv4.conf.default.accept_redirects` to 0.
 - Add or modify the line `net.ipv4.conf.default.accept_redirects = 0` in a sysctl configuration file (e.g., `/etc/sysctl.d/99-network.conf`).
 - Run `sudo sysctl -p` to apply.

Rule 3.3.1.10: Ensure `net.ipv4.conf.all.secure_redirects` is configured

- **Status:** FAIL
- **Evidence:** The `net.ipv4.conf.all.secure_redirects` parameter is set to 1 at runtime, but the recommended value is 0. It is also not persisted.

- `network_config.sysctl_ipv4.net.ipv4.conf.all.secure_redirects.r`
1
- `network_config.sysctl_ipv4.net.ipv4.conf.all.secure_redirects.p`
null

- **Recommendation:** Set `net.ipv4.conf.all.secure_redirects` to 0.

- Add or modify the line `net.ipv4.conf.all.secure_redirects = 0` in a `sysctl` configuration file (e.g., `/etc/sysctl.d/99-network.conf`).
- Run `sudo sysctl -p` to apply.

Rule 3.3.1.11: Ensure `net.ipv4.conf.default.secure_redirects` is configured

- **Status:** FAIL
- **Evidence:** The `net.ipv4.conf.default.secure_redirects` parameter is set to 1 at runtime, but the recommended value is 0. It is also not persisted.
 - `network_config.sysctl_ipv4.net.ipv4.conf.default.secure_redirec`
1
 - `network_config.sysctl_ipv4.net.ipv4.conf.default.secure_redirec`
null
- **Recommendation:** Set `net.ipv4.conf.default.secure_redirects` to 0.
 - Add or modify the line `net.ipv4.conf.default.secure_redirects = 0` in a `sysctl` configuration file (e.g., `/etc/sysctl.d/99-network.conf`).
 - Run `sudo sysctl -p` to apply.

Rule 3.3.1.12: Ensure `net.ipv4.conf.all.rp_filter` is configured

- **Status:** PASS
- **Evidence:** The `net.ipv4.conf.all.rp_filter` parameter is set to 2 at runtime and is persisted in `/etc/sysctl.d/10-network-security.conf`. Both 1 and 2 are compliant.
 - `network_config.sysctl_ipv4.net.ipv4.conf.all.rp_filter.runtime_`
2
 - `network_config.sysctl_ipv4.net.ipv4.conf.all.rp_filter.persiste`
2
 - `network_config.sysctl_ipv4.net.ipv4.conf.all.rp_filter.persiste`
`/etc/sysctl.d/10-network-security.conf`
- **Recommendation:** None.

Rule 3.3.1.13: Ensure `net.ipv4.conf.default.rp_filter` is configured

- **Status:** PASS
- **Evidence:** The `net.ipv4.conf.default.rp_filter` parameter is set to 2 at runtime and is persisted in `/etc/sysctl.d/10-network-security.conf`. Both 1 and 2 are compliant.
 - `network_config.sysctl_ipv4.net.ipv4.conf.default.rp_filter.runt`
2

- `network_config.sysctl_ipv4.net.ipv4.conf.default.rp_filter.pers`
2
- `network_config.sysctl_ipv4.net.ipv4.conf.default.rp_filter.pers`
/etc/sysctl.d/10-network-security.conf

- **Recommendation:** None.

Rule 3.3.1.14: Ensure net.ipv4.conf.all.accept_source_route is configured

- **Status:** PASS
- **Evidence:** The `net.ipv4.conf.all.accept_source_route` parameter is set to 0 at runtime and is not explicitly persisted (default is 0).

- `network_config.sysctl_ipv4.net.ipv4.conf.all.accept_source_rout`
0
- `network_config.sysctl_ipv4.net.ipv4.conf.all.accept_source_rout`
null

- **Recommendation:** None.

Rule 3.3.1.15: Ensure net.ipv4.conf.default.accept_source_route is configured

- **Status:** FAIL
- **Evidence:** The `net.ipv4.conf.default.accept_source_route` parameter is set to 1 at runtime, but the recommended value is 0. It is also not persisted.

- `network_config.sysctl_ipv4.net.ipv4.conf.default.accept_source_`
1
- `network_config.sysctl_ipv4.net.ipv4.conf.default.accept_source_`
null

- **Recommendation:** Set `net.ipv4.conf.default.accept_source_route` to 0.
 - Add or modify the line `net.ipv4.conf.default.accept_source_route = 0` in a sysctl configuration file (e.g., `/etc/sysctl.d/99-network.conf`).
 - Run `sudo sysctl -p` to apply.

Rule 3.3.1.16: Ensure net.ipv4.conf.all.log_martians is configured

- **Status:** FAIL
- **Evidence:** The `net.ipv4.conf.all.log_martians` parameter is set to 0 at runtime, but the recommended value is 1. It is also not persisted.

- `network_config.sysctl_ipv4.net.ipv4.conf.all.log_martians.runti`
0
- `network_config.sysctl_ipv4.net.ipv4.conf.all.log_martians.persi`
null

- **Recommendation:** Set `net.ipv4.conf.all.log_martians` to 1.
 - Add or modify the line `net.ipv4.conf.all.log_martians = 1` in a sysctl configuration file (e.g., `/etc/sysctl.d/99-network.conf`).
 - Run `sudo sysctl -p` to apply.

Rule 3.3.1.17: Ensure net.ipv4.conf.default.log_martians is configured

- **Status:** FAIL
- **Evidence:** The `net.ipv4.conf.default.log_martians` parameter is set to 0 at runtime, but the recommended value is 1. It is also not persisted.
 - `network_config.sysctl_ipv4.net.ipv4.conf.default.log_martians.r`
0
 - `network_config.sysctl_ipv4.net.ipv4.conf.default.log_martians.p`
null
- **Recommendation:** Set `net.ipv4.conf.default.log_martians` to 1.
 - Add or modify the line `net.ipv4.conf.default.log_martians = 1` in a `sysctl` configuration file (e.g., `/etc/sysctl.d/99-network.conf`).
 - Run `sudo sysctl -p` to apply.

Rule 3.3.1.18: Ensure net.ipv4.tcp_syncookies is configured

- **Status:** PASS
- **Evidence:** The `net.ipv4.tcp_syncookies` parameter is set to 1 at runtime and is not explicitly persisted (default is 1).
 - `network_config.sysctl_ipv4.net.ipv4.tcp_syncookies.runtime_valu`
1
 - `network_config.sysctl_ipv4.net.ipv4.tcp_syncookies.persisted_va`
null
- **Recommendation:** None.

Rule 3.3.2.1: Ensure net.ipv6.conf.all.forwarding is configured

- **Status:** PASS
- **Evidence:** The `net.ipv6.conf.all.forwarding` parameter is set to 0 at runtime and is not explicitly persisted (default is 0).
 - `network_config.sysctl_ipv6.net.ipv6.conf.all.forwarding.runtime`
0
 - `network_config.sysctl_ipv6.net.ipv6.conf.all.forwarding.persist`
null
- **Recommendation:** None.

Rule 3.3.2.2: Ensure net.ipv6.conf.default.forwarding is configured

- **Status:** PASS
- **Evidence:** The `net.ipv6.conf.default.forwarding` parameter is set to 0 at runtime and is not explicitly persisted (default is 0).
 - `network_config.sysctl_ipv6.net.ipv6.conf.default.forwarding.rur`
0
 - `network_config.sysctl_ipv6.net.ipv6.conf.default.forwarding.per`
null

- **Recommendation:** None.

Rule 3.3.2.3: Ensure net.ipv6.conf.all.accept_redirects is configured

- **Status:** FAIL
- **Evidence:** The `net.ipv6.conf.all.accept_redirects` parameter is set to 1 at runtime, but the recommended value is 0. It is also not persisted.
 - `network_config.sysctl_ipv6.net.ipv6.conf.all.accept_redirects.r`
1
 - `network_config.sysctl_ipv6.net.ipv6.conf.all.accept_redirects.p`
null
- **Recommendation:** Set `net.ipv6.conf.all.accept_redirects` to 0.
 - Add or modify the line `net.ipv6.conf.all.accept_redirects = 0` in a sysctl configuration file (e.g., `/etc/sysctl.d/99-network.conf`).
 - Run `sudo sysctl -p` to apply.

Rule 3.3.2.4: Ensure net.ipv6.conf.default.accept_redirects is configured

- **Status:** FAIL
- **Evidence:** The `net.ipv6.conf.default.accept_redirects` parameter is set to 1 at runtime, but the recommended value is 0. It is also not persisted.
 - `network_config.sysctl_ipv6.net.ipv6.conf.default.accept_redirec`
1
 - `network_config.sysctl_ipv6.net.ipv6.conf.default.accept_redirec`
null
- **Recommendation:** Set `net.ipv6.conf.default.accept_redirects` to 0.
 - Add or modify the line `net.ipv6.conf.default.accept_redirects = 0` in a sysctl configuration file (e.g., `/etc/sysctl.d/99-network.conf`).
 - Run `sudo sysctl -p` to apply.

Rule 3.3.2.5: Ensure net.ipv6.conf.all.accept_source_route is configured

- **Status:** PASS
- **Evidence:** The `net.ipv6.conf.all.accept_source_route` parameter is set to 0 at runtime and is not explicitly persisted (default is 0).
 - `network_config.sysctl_ipv6.net.ipv6.conf.all.accept_source_rout`
0
 - `network_config.sysctl_ipv6.net.ipv6.conf.all.accept_source_rout`
null
- **Recommendation:** None.

Rule 3.3.2.6: Ensure net.ipv6.conf.default.accept_source_route is configured

- **Status:** PASS

- **Evidence:** The `net.ipv6.conf.default.accept_source_route` parameter is set to 0 at runtime and is not explicitly persisted (default is 0).
 - `network_config.sysctl_ipv6.net.ipv6.conf.default.accept_source_0`
 - `network_config.sysctl_ipv6.net.ipv6.conf.default.accept_source_null`
- **Recommendation:** None.

Rule 3.3.2.7: Ensure `net.ipv6.conf.all.accept_ra` is configured

- **Status:** FAIL
- **Evidence:** The `net.ipv6.conf.all.accept_ra` parameter is set to 1 at runtime, but the recommended value is 0. It is also not persisted.
 - `network_config.sysctl_ipv6.net.ipv6.conf.all.accept_ra.runtime_1`
 - `network_config.sysctl_ipv6.net.ipv6.conf.all.accept_ra.persiste null`
- **Recommendation:** Set `net.ipv6.conf.all.accept_ra` to 0.
 - Add or modify the line `net.ipv6.conf.all.accept_ra = 0` in a `sysctl` configuration file (e.g., `/etc/sysctl.d/99-network.conf`).
 - Run `sudo sysctl -p` to apply.

Rule 3.3.2.8: Ensure `net.ipv6.conf.default.accept_ra` is configured

- **Status:** FAIL
- **Evidence:** The `net.ipv6.conf.default.accept_ra` parameter is set to 1 at runtime, but the recommended value is 0. It is also not persisted.
 - `network_config.sysctl_ipv6.net.ipv6.conf.default.accept_ra.runt 1`
 - `network_config.sysctl_ipv6.net.ipv6.conf.default.accept_ra.pers null`
- **Recommendation:** Set `net.ipv6.conf.default.accept_ra` to 0.
 - Add or modify the line `net.ipv6.conf.default.accept_ra = 0` in a `sysctl` configuration file (e.g., `/etc/sysctl.d/99-network.conf`).
 - Run `sudo sysctl -p` to apply.

Section 4.1 – Configure Uncomplicated Firewall

Rule 4.1.1: Ensure `ufw` is installed

- **Status:** PASS
- **Evidence:** The `ufw` package is installed.
 - `ufw_firewall.ufw_installed:true`
 - `ufw_firewall.ufw_dpkg_status:install ok installed`

- **Recommendation:** None.

Rule 4.1.2: Ensure ufw service is configured

- **Status:** FAIL
- **Evidence:** The ufw service is enabled but currently inactive.
 - `ufw_firewall.service_configured.is_enabled: enabled`
 - `ufw_firewall.service_configured.is_active: active` (Note: `status_verbos` shows inactive, which is contradictory. Assuming `is_active` is the primary indicator for service status, but the `status_verbos` indicates it's not actually running.)
 - `ufw_firewall.service_configured.status_verbos: Status: inactive`
- **Recommendation:** Enable and start the UFW firewall.
 1. `sudo ufw enable`
 2. Verify status: `sudo ufw status verbose`

Rule 4.1.3: Ensure ufw incoming default is configured

- **Status:** PASS
- **Evidence:** The default incoming policy for UFW is set to DROP in `/etc/default/ufw`.
 - `ufw_firewall.default_policies.etc_default_ufw.DEFAULT_INPUT_POI`
DROP
- **Recommendation:** None.

Rule 4.1.4: Ensure ufw outgoing default is configured

- **Status:** PASS
- **Evidence:** The default outgoing policy for UFW is set to ACCEPT in `/etc/default/ufw`.
 - `ufw_firewall.default_policies.etc_default_ufw.DEFAULT_OUTPUT_PC`
ACCEPT
- **Recommendation:** None.

Rule 4.1.5: Ensure ufw routed default is configured

- **Status:** PASS
- **Evidence:** The default routed policy for UFW is set to DROP in `/etc/default/ufw`.
 - `ufw_firewall.default_policies.etc_default_ufw.DEFAULT_FORWARD_F`
DROP
- **Recommendation:** None.

Section 5.3 – Pluggable Authentication Modules (PAM)

Rule 5.3.1.1: Ensure latest version of pam is installed

- **Status:** PASS
- **Evidence:** The `libpam-runtime` package is installed and is at the latest available version.
 - `pam.pam_packages[0].installed: true`
 - `pam.pam_packages[0].installed_version: 1.5.3-5ubuntu5.5`
 - `pam.pam_packages[0].candidate_version: 1.5.3-5ubuntu5.5`
- **Recommendation:** None.

Rule 5.3.1.2: Ensure latest version of libpam-modules is installed

- **Status:** PASS
- **Evidence:** The `libpam-modules` package is installed and is at the latest available version.
 - `pam.pam_packages[1].installed: true`
 - `pam.pam_packages[1].installed_version: 1.5.3-5ubuntu5.5`
 - `pam.pam_packages[1].candidate_version: 1.5.3-5ubuntu5.5`
- **Recommendation:** None.

Rule 5.3.1.3: Ensure latest version of libpam-pwquality is installed

- **Status:** FAIL
- **Evidence:** The `libpam-pwquality` package is not installed.
 - `pam.pam_packages[2].installed: false`
- **Recommendation:** Install the `libpam-pwquality` package: `sudo apt install libpam-pwquality`.

Rule 5.3.1.4: Ensure latest version of cracklib-runtime is installed

- **Status:** FAIL
- **Evidence:** The `cracklib-runtime` package is not installed.
 - `pam.pam_packages[3].installed: false`
- **Recommendation:** Install the `cracklib-runtime` package: `sudo apt install cracklib-runtime`.

Rule 5.3.2.1: Ensure pam_unix module is enabled

- **Status:** PASS
- **Evidence:** The `pam_unix.so` module is enabled and configured in `common-auth`, `common-account`, `common-password`, and `common-session`.
 - `pam.pam_profiles_raw.module_lines.pam_unix.so: Contains active lines.`
- **Recommendation:** None.

Rule 5.3.2.2: Ensure pam_faillock module is enabled

- **Status:** FAIL
- **Evidence:** The `pam_faillock.so` module is not enabled in any PAM configuration files.
 - `pam.pam_profiles_raw.module_lines.pam_faillock.so: []`
- **Recommendation:** Enable the `pam_faillock.so` module in the PAM configuration.
 1. Edit `/etc/pam.d/common-auth` and add the following lines at the top of the file:

```
auth required pam_faillock.so preauth silent deny=5 unlock_time
auth [success=1 default=ignore] pam_unix.so nullok
```
 2. Edit `/etc/pam.d/common-account` and add the following line at the top of the file:

```
account required pam_faillock.so
```
 3. Run `sudo pam-auth-update` and ensure faillock is selected.

Rule 5.3.2.3: Ensure pam_pwquality module is enabled

- **Status:** FAIL
- **Evidence:** The `pam_pwquality.so` module is not enabled in any PAM configuration files.
 - `pam.pam_profiles_raw.module_lines.pam_pwquality.so: []`
- **Recommendation:** Enable the `pam_pwquality.so` module in the PAM configuration.
 1. Install `libpam-pwquality` (if not already done per 5.3.1.3).
 2. Edit `/etc/pam.d/common-password` and add `password requisite pam_pwquality.so retry=3` before `pam_unix.so`.
 3. Run `sudo pam-auth-update` and ensure pwquality is selected.

Rule 5.3.2.4: Ensure pam_pwhistory module is enabled

- **Status:** FAIL
- **Evidence:** The `pam_pwhistory.so` module is not enabled in any PAM configuration files.
 - `pam.pam_profiles_raw.module_lines.pam_pwhistory.so: []`
- **Recommendation:** Enable the `pam_pwhistory.so` module in the PAM configuration.
 1. Edit `/etc/pam.d/common-password` and add `password required pam_pwhistory.so remember=24 use_authtok` after `pam_pwquality.so` (if enabled) or before `pam_unix.so`.
 2. Run `sudo pam-auth-update` and ensure pwhistory is selected.

Rule 5.3.3.1.1: Ensure password failed attempts lockout is configured

- **Status:** FAIL
- **Evidence:** The `pam_faillock` module is not enabled, so password failed attempts lockout is not configured.
 - `pam.pam_faillock.faillock_conf_settings.deny: null`
 - `pam.pam_faillock.faillock_inline_pam_settings.deny: null`
- **Recommendation:** Enable `pam_faillock` and configure `deny=5` (or less) in `/etc/security/faillock.conf` or inline PAM arguments. (See 5.3.2.2 for enabling `pam_faillock`).

Rule 5.3.3.1.2: Ensure password unlock time is configured

- **Status:** FAIL
- **Evidence:** The `pam_faillock` module is not enabled, so password unlock time is not configured.
 - `pam.pam_faillock.faillock_conf_settings.unlock_time: null`
 - `pam.pam_faillock.faillock_inline_pam_settings.unlock_time: null`
- **Recommendation:** Enable `pam_faillock` and configure `unlock_time=900` (or greater, or 0 for never) in `/etc/security/faillock.conf` or inline PAM arguments. (See 5.3.2.2 for enabling `pam_faillock`).

Rule 5.3.3.1.3: Ensure password failed attempts lockout includes root account

- **Status:** FAIL
- **Evidence:** The `pam_faillock` module is not enabled, so root account lockout is not configured.
 - `pam.pam_faillock.faillock_conf_settings.even_deny_root: null`
 - `pam.pam_faillock.faillock_inline_pam_settings.even_deny_root: null`
- **Recommendation:** Enable `pam_faillock` and include `even_deny_root` in `/etc/security/faillock.conf` or inline PAM arguments. (See 5.3.2.2 for enabling `pam_faillock`).

Rule 5.3.3.2.1: Ensure password number of changed characters is configured

- **Status:** FAIL
- **Evidence:** The `pam_pwquality` module is not enabled, so `difok` is not configured.
 - `pam.pam_pwquality.pwquality_conf_settings.difok: null`
 - `pam.pam_pwquality.pwquality_inline_pam_settings.difok: null`
- **Recommendation:** Enable `pam_pwquality` and configure `difok=2` (or greater) in `/etc/security/pwquality.conf` or inline PAM arguments. (See 5.3.2.3 for enabling `pam_pwquality`).

Rule 5.3.3.2.2: Ensure password length is configured

- **Status:** FAIL
- **Evidence:** The `pam_pwquality` module is not enabled, so `minlen` is not configured.
 - `pam.pam_pwquality.pwquality_conf_settings.minlen: null`
 - `pam.pam_pwquality.pwquality_inline_pam_settings.minlen: null`
- **Recommendation:** Enable `pam_pwquality` and configure `minlen=14` (or greater) in `/etc/security/pwquality.conf` or inline PAM arguments. (See 5.3.2.3 for enabling `pam_pwquality`).

Rule 5.3.3.2.3: Ensure password complexity is configured

- **Status:** INFORMATIONAL
- **Evidence:** The `pam_pwquality` module is not enabled, so specific complexity requirements (`dcredit`, `ucredit`, `lcredit`, `ocredit`) are not configured.
 - `pam.pam_pwquality.pwquality_conf_settings: All complexity settings are null.`
- **Recommendation:** Enable `pam_pwquality` and configure `dcredit`, `ucredit`, `lcredit`, `ocredit` in `/etc/security/pwquality.conf` or inline PAM arguments to enforce password complexity. (See 5.3.2.3 for enabling `pam_pwquality`).

Rule 5.3.3.2.4: Ensure password same consecutive characters is configured

- **Status:** FAIL
- **Evidence:** The `pam_pwquality` module is not enabled, so `maxrepeat` is not configured.
 - `pam.pam_pwquality.pwquality_conf_settings.maxrepeat: null`
 - `pam.pam_pwquality.pwquality_inline_pam_settings.maxrepeat: null`
- **Recommendation:** Enable `pam_pwquality` and configure `maxrepeat=3` (or less) in `/etc/security/pwquality.conf` or inline PAM arguments. (See 5.3.2.3 for enabling `pam_pwquality`).

Rule 5.3.3.2.5: Ensure password maximum sequential characters is configured

- **Status:** FAIL
- **Evidence:** The `pam_pwquality` module is not enabled, so `maxsequence` is not configured.
 - `pam.pam_pwquality.pwquality_conf_settings.maxsequence: null`
 - `pam.pam_pwquality.pwquality_inline_pam_settings.maxsequence: null`
- **Recommendation:** Enable `pam_pwquality` and configure `maxsequence=3` (or less) in `/etc/security/pwquality.conf` or inline PAM arguments. (See 5.3.2.3 for

enabling pam_pwquality).

Rule 5.3.3.2.6: Ensure password dictionary check is enabled

- **Status:** FAIL
- **Evidence:** The pam_pwquality module is not enabled, so dictcheck is not configured.
 - pam.pam_pwquality.pwquality_conf_settings.dictcheck: null
 - pam.pam_pwquality.pwquality_inline_pam_settings.dictcheck: null
- **Recommendation:** Enable pam_pwquality and ensure dictcheck=1 (default) in /etc/security/pwquality.conf or inline PAM arguments. (See 5.3.2.3 for enabling pam_pwquality).

Rule 5.3.3.2.7: Ensure password quality checking is enforced

- **Status:** FAIL
- **Evidence:** The pam_pwquality module is not enabled, so password quality checking is not enforced.
 - pam.pam_pwquality.pwquality_conf_settings.enforcing: null
 - pam.pam_pwquality.pwquality_inline_pam_settings.enforcing: null
- **Recommendation:** Enable pam_pwquality and ensure enforcing=1 (default) in /etc/security/pwquality.conf or inline PAM arguments. (See 5.3.2.3 for enabling pam_pwquality).

Rule 5.3.3.2.8: Ensure password quality is enforced for the root user

- **Status:** FAIL
- **Evidence:** The pam_pwquality module is not enabled, so password quality enforcement for root is not configured.
 - pam.pam_pwquality.pwquality_conf_settings.enforce_for_root: null
 - pam.pam_pwquality.pwquality_inline_pam_settings.enforce_for_root: null
- **Recommendation:** Enable pam_pwquality and include enforce_for_root in /etc/security/pwquality.conf or inline PAM arguments. (See 5.3.2.3 for enabling pam_pwquality).

Rule 5.3.3.3.1: Ensure password history remember is configured

- **Status:** FAIL
- **Evidence:** The pam_pwhistory module is not enabled, so password history is not configured.
 - pam.pam_pwhistory.pwhistory_inline_pam_settings.remember: null
- **Recommendation:** Enable pam_pwhistory and configure remember=24 (or greater)

in the PAM arguments for `pam_pwhistory.so`. (See 5.3.2.4 for enabling `pam_pwhistory`).

Rule 5.3.3.3.2: Ensure password history is enforced for the root user

- **Status:** FAIL
- **Evidence:** The `pam_pwhistory` module is not enabled, so password history enforcement for root is not configured.
 - `pam.pam_pwhistory.pwhistory_inline_pam_settings.enforce_for_root null`
- **Recommendation:** Enable `pam_pwhistory` and include `enforce_for_root` in the PAM arguments for `pam_pwhistory.so`. (See 5.3.2.4 for enabling `pam_pwhistory`).

Rule 5.3.3.3.3: Ensure `pam_pwhistory` includes `use_authok`

- **Status:** FAIL
- **Evidence:** The `pam_pwhistory` module is not enabled, so `use_authok` is not configured.
 - `pam.pam_pwhistory.pwhistory_inline_pam_settings.use_authok: null`
- **Recommendation:** Enable `pam_pwhistory` and include `use_authok` in the PAM arguments for `pam_pwhistory.so`. (See 5.3.2.4 for enabling `pam_pwhistory`).

Rule 5.3.3.4.1: Ensure `pam_unix` does not include `nullok`

- **Status:** FAIL
- **Evidence:** The `pam_unix.so` module in `/etc/pam.d/common-auth` includes the `nullok` option.
 - `pam.pam_unix.pam_unix_flag_presence.nullok: present`
 - `pam.pam_unix.pam_unix_lines[0].line: auth [success=1 default=ignore] pam_unix.so nullok`
- **Recommendation:** Remove the `nullok` option from the `pam_unix.so` line in `/etc/pam.d/common-auth`. This prevents users from authenticating with empty passwords.

Rule 5.3.3.4.2: Ensure `pam_unix` does not include `remember`

- **Status:** PASS
- **Evidence:** The `pam_unix.so` module does not include the `remember` option.
 - `pam.pam_unix.pam_unix_flag_presence.remember: null`
- **Recommendation:** None.

Rule 5.3.3.4.3: Ensure `pam_unix` includes a strong password hashing algorithm

- **Status:** PASS

- **Evidence:** The `pam_unix.so` module in `/etc/pam.d/common-password` includes the `yescrypt` option, which is a strong hashing algorithm.
 - `pam.pam_unix.pam_unix_flag_presence.yescrypt: present`
 - `pam.pam_unix.pam_unix_lines[2].line: password [success=1 default=ignore] pam_unix.so obscure yescrypt`
- **Recommendation:** None.

Rule 5.3.3.4.4: Ensure pam_unix includes use_authtok

- **Status:** FAIL
- **Evidence:** The `pam_unix.so` module in `/etc/pam.d/common-password` does not include the `use_authtok` option.
 - `pam.pam_unix.pam_unix_flag_presence.use_authtok: null`
 - `pam.pam_unix.pam_unix_lines[2].line: password [success=1 default=ignore] pam_unix.so obscure yescrypt`
- **Recommendation:** Add the `use_authtok` option to the `pam_unix.so` line in `/etc/pam.d/common-password`. This ensures that the new password is used for all modules in the password stack.

Section 6.1 – System Logging

Rule 6.1.1.1.1: Ensure journald service is active

- **Status:** PASS
- **Evidence:** The `systemd-journald.service` is active.
 - `system_logging.journald.systemd-journald_active: active`
- **Recommendation:** None.

Rule 6.1.1.1.2: Ensure systemd-journal-remote service is not in use

- **Status:** PASS
- **Evidence:** The `systemd-journal-remote` package is not installed, and `systemd-journal-upload` and `systemd-journal-remote.socket` services are inactive.
 - `system_logging.journald.systemd-journal-remote_installed: false`
 - `system_logging.journald.systemd-journal-upload_active: inactive`
 - `system_logging.journald.systemd-journal-remote_socket_active: inactive`
- **Recommendation:** None.

Rule 6.1.1.1.3: Ensure journald is configured to send logs to rsyslog

- **Status:** FAIL
- **Evidence:** The `ForwardToSyslog` option is commented out in `/etc/systemd/`

journald.conf, indicating that logs are not explicitly forwarded to rsyslog.

- system_logging.journald.journald_configs[0].content:
#ForwardToSyslog=no

- **Recommendation:** Uncomment and set ForwardToSyslog=yes in /etc/systemd/journald.conf to ensure journald logs are forwarded to rsyslog for centralized management. Then restart systemd-journald.service.

Rule 6.1.1.1.4: Ensure journald log file access is configured

- **Status:** INFORMATIONAL
- **Evidence:** The /run/log/journal and /var/log/journal directories have mode 2755, owner root, group systemd-journal. Log files within /var/log/journal have mode 0640, owner root, group systemd-journal.
 - /run/log/journal: mode=2755, owner=root, group=999 (systemd-journal)
 - /var/log/journal: mode=2755, owner=root, group=999 (systemd-journal)
 - **Example log file:** /var/log/journal/196b6265ff2e44169e0a73c65801746c/system.journal: mode=0640, owner=root, group=999 (systemd-journal)
- **Recommendation:** Ensure that the systemd-journal group (GID 999) has appropriate access and that no unauthorized users are part of this group.

Rule 6.1.1.1.5: Ensure journald log file rotation is configured

- **Status:** INFORMATIONAL
- **Evidence:** The MaxRetentionSec and MaxFileSec options are commented out in /etc/systemd/journald.conf, indicating default rotation settings are in use.
 - system_logging.journald.journald_configs[0].content:
#MaxRetentionSec=, #MaxFileSec=1month
- **Recommendation:** Explicitly configure MaxRetentionSec and MaxFileSec in /etc/systemd/journald.conf to meet organizational log retention policies. For example, MaxRetentionSec=1year and MaxFileSec=1month.

Rule 6.1.1.1.6: Ensure journald Storage is configured

- **Status:** FAIL
- **Evidence:** The Storage option is commented out in /etc/systemd/journald.conf, defaulting to auto. While /var/log/journal exists, explicit configuration to persistent is recommended.
 - system_logging.journald.journald_configs[0].content:
#Storage=auto
 - system_logging.journald.journald_log_dirs: Includes /var/log/journal
- **Recommendation:** Uncomment and set Storage=persistent in /etc/systemd/journald.conf to ensure journal logs are stored persistently across reboots. Then restart systemd-journald.service.

Rule 6.1.1.1.7: Ensure journald Compress is configured

- **Status:** FAIL
- **Evidence:** The Compress option is commented out in `/etc/systemd/journald.conf`, defaulting to yes. Explicit configuration is recommended.
 - `system_logging.journald.journald_configs[0].content:`
`#Compress=yes`
- **Recommendation:** Uncomment and set `Compress=yes` in `/etc/systemd/journald.conf` to ensure journal logs are compressed. Then restart `systemd-journald.service`.

Rule 6.1.2.1: Ensure rsyslog is installed

- **Status:** PASS
- **Evidence:** The rsyslog package is installed.
 - `system_logging.rsyslog.rsyslog_installed: true`
- **Recommendation:** None.

Rule 6.1.2.2: Ensure rsyslog service is enabled and active

- **Status:** PASS
- **Evidence:** The `rsyslog.service` is enabled and active.
 - `system_logging.rsyslog.rsyslog_enabled: enabled`
 - `system_logging.rsyslog.rsyslog_active: active`
- **Recommendation:** None.

Rule 6.1.2.3: Ensure rsyslog log file creation mode is configured

- **Status:** PASS
- **Evidence:** The `$FileCreateMode` is set to `0640` in `/etc/rsyslog.conf`.
 - `system_logging.rsyslog.rsyslog_configs[0].content:`
`$FileCreateMode 0640`
- **Recommendation:** None.

Rule 6.1.2.4: Ensure rsyslog logging is configured

- **Status:** INFORMATIONAL
- **Evidence:** Rsyslog is configured to log `auth, authpriv.*` to `/var/log/auth.log` and `*.*;auth,authpriv.none` to `/var/log/syslog`, among other default rules in `/etc/rsyslog.d/50-default.conf`.
- **Recommendation:** Review the rsyslog configuration files (`/etc/rsyslog.conf` and `/etc/rsyslog.d/*.conf`) to ensure all necessary log facilities and priorities are captured and directed to appropriate files according to organizational logging policies.

Rule 6.1.2.5: Ensure rsyslog is configured to send logs to a remote log

host

- **Status:** INFORMATIONAL
- **Evidence:** The rsyslog configuration files do not explicitly show rules for forwarding logs to a remote log host. The `imudp` and `imtcp` modules are commented out, suggesting no remote reception, but also no forwarding.
 - `system_logging.rsyslog.rsyslog_configs`: No `$ActionFileDefaultTemplate` or @@ forwarding rules found.
- **Recommendation:** Configure rsyslog to send logs to a centralized, remote log host for enhanced security and retention. Add a line like `*.* @@remote-log-host:514` (for UDP) or `*.* @remote-log-host:514` (for TCP) to `/etc/rsyslog.conf` or a file in `/etc/rsyslog.d/`.

Rule 6.1.2.6: Ensure rsyslog is not configured to receive logs from a remote client

- **Status:** PASS
- **Evidence:** The `imudp` and `imtcp` modules for receiving remote syslog messages are commented out in `/etc/rsyslog.conf`.

- `system_logging.rsyslog.rsyslog_configs[0].content`:
`#module(load="imudp"), #input(type="imudp" port="514"),`
`#module(load="imtcp"), #input(type="imtcp" port="514")`

- **Recommendation:** None.

Rule 6.1.2.7: Ensure logrotate is configured

- **Status:** INFORMATIONAL
- **Evidence:** `logrotate` is installed and configured with a weekly rotation, 4 weeks of backlogs, and includes configurations from `/etc/logrotate.d/`. The `logrotate.timer` is enabled and active.

- `system_logging.rsyslog.logrotate_configs[0].content`: `weekly,`
`rotate 4, create, include /etc/logrotate.d`
- `system_logging.rsyslog.logrotate_timer_enabled`: `enabled`
- `system_logging.rsyslog.logrotate_timer_active`: `active`

- **Recommendation:** Review individual `logrotate` configurations in `/etc/logrotate.d/` to ensure they meet specific retention and compression requirements for all critical log files.

Rule 6.1.2.8: Ensure rsyslog-gnutls is installed

- **Status:** FAIL
- **Evidence:** The `rsyslog-gnutls` package is not installed.

- `system_logging.rsyslog.rsyslog_gnutls_installed`: `false`

- **Recommendation:** Install the `rsyslog-gnutls` package: `sudo apt install rsyslog-gnutls`. This is required for secure remote log forwarding.

Rule 6.1.2.9: Ensure rsyslog forwarding uses gtls

- **Status:** FAIL
- **Evidence:** The `rsyslog-gnutls` package is not installed, so secure forwarding using GTLS cannot be configured or verified.

○ `system_logging.rsyslog.rsyslog_gnutls_installed: false`

- **Recommendation:** Install `rsyslog-gnutls` and configure `rsyslog` to use GTLS for remote log forwarding. (See 6.1.2.8 and 6.1.2.5).

Rule 6.1.2.10: Ensure rsyslog CA certificates are configured

- **Status:** INFORMATIONAL
- **Evidence:** The `rsyslog-gnutls` package is not installed, so CA certificates for secure remote logging are not configured or verifiable.

○ `system_logging.rsyslog.rsyslog_gnutls_installed: false`

- **Recommendation:** Install `rsyslog-gnutls` and configure appropriate CA certificates and authentication modes (e.g., `StreamDriverAuthMode="x509/certvalid"`) for secure remote log forwarding.

Rule 6.1.3.1: Ensure access to all logfiles has been configured

- **Status:** FAIL
- **Evidence:** Several log files have permissions that are too permissive or incorrect.

○ `/var/log/wtmp: 0664 (should be 0600 or 0640)`
○ `/var/log/btmp: 0660 (should be 0600 or 0640)`
○ `/var/log/lastlog: 0664 (should be 0600 or 0640)`
○ `/var/log/faillog: 0644 (should be 0600 or 0640)`
○ `/var/log/apt/history.log: 0644 (should be 0640)`
○ `/var/log/unattended-upgrades/unattended-upgrades.log: 0644 (should be 0640)`
○ `/var/log/unattended-upgrades/unattended-upgrades-shutdown.log: 0644 (should be 0640)`

- **Recommendation:** Adjust permissions for the identified log files to be more restrictive. For example:

○ `sudo chmod 0600 /var/log/wtmp`
○ `sudo chmod 0600 /var/log/btmp`
○ `sudo chmod 0600 /var/log/lastlog`
○ `sudo chmod 0600 /var/log/faillog`
○ `sudo chmod 0640 /var/log/apt/history.log`
○ `sudo chmod 0640 /var/log/unattended-upgrades/unattended-upgrades.log`
○ `sudo chmod 0640 /var/log/unattended-upgrades/unattended-upgrades-shutdown.log` Also, ensure `logrotate` configurations for these files enforce the correct permissions upon rotation.

Section 6.2 – System Auditing (auditd)

Rule 6.2.1.1: Ensure auditd packages are installed

- **Status:** FAIL
- **Evidence:** The auditd package is not installed on the system.

○ system_auditing.auditd_installed: false

- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.1.2: Ensure auditd service is enabled and active

- **Status:** FAIL
- **Evidence:** The auditd package is not installed on the system.

○ system_auditing.auditd_installed: false

- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.1.3: Ensure auditing for processes that start prior to auditd is enabled

- **Status:** FAIL
- **Evidence:** The auditd package is not installed on the system.

○ system_auditing.auditd_installed: false

- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.1.4: Ensure audit_backlog_limit is configured

- **Status:** FAIL
- **Evidence:** The auditd package is not installed on the system.

○ system_auditing.auditd_installed: false

- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.2.1: Ensure audit log storage size is configured

- **Status:** FAIL
- **Evidence:** The auditd package is not installed on the system.

○ system_auditing.auditd_installed: false

- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.2.2: Ensure audit logs are not automatically deleted

- **Status:** FAIL
- **Evidence:** The auditd package is not installed on the system.
 - `system_auditing.auditd_installed: false`
- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.2.3: Ensure system is disabled when audit logs are full

- **Status:** FAIL
- **Evidence:** The auditd package is not installed on the system.
 - `system_auditing.auditd_installed: false`
- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.2.4: Ensure system warns when audit logs are low on space

- **Status:** FAIL
- **Evidence:** The auditd package is not installed on the system.
 - `system_auditing.auditd_installed: false`
- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.3.1: Ensure changes to system administration scope (sudoers) is collected

- **Status:** FAIL
- **Evidence:** The auditd package is not installed on the system.
 - `system_auditing.auditd_installed: false`
- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.3.2: Ensure actions as another user are always logged

- **Status:** FAIL
- **Evidence:** The auditd package is not installed on the system.
 - `system_auditing.auditd_installed: false`
- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.3.3: Ensure events that modify the sudo log file are collected

- **Status:** FAIL
- **Evidence:** The auditd package is not installed on the system.

○ `system_auditing.auditd_installed: false`

- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.3.4: Ensure events that modify date and time information are collected

- **Status:** FAIL
- **Evidence:** The auditd package is not installed on the system.

○ `system_auditing.auditd_installed: false`

- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.3.5: Ensure events that modify sethostname and setdomainname are collected

- **Status:** FAIL
- **Evidence:** The auditd package is not installed on the system.

○ `system_auditing.auditd_installed: false`

- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.3.6: Ensure events that modify /etc/issue and /etc/issue.net are collected

- **Status:** FAIL
- **Evidence:** The auditd package is not installed on the system.

○ `system_auditing.auditd_installed: false`

- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.3.7: Ensure events that modify /etc/hosts and /etc/hostname are collected

- **Status:** FAIL
- **Evidence:** The auditd package is not installed on the system.

○ `system_auditing.auditd_installed: false`

- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.3.8: Ensure events that modify /etc/network and /etc/networks are collected

- **Status:** FAIL
- **Evidence:** The auditd package is not installed on the system.

○ system_auditing.auditd_installed: false

- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.3.9: Ensure events that modify /etc/netplan are collected

- **Status:** FAIL
- **Evidence:** The auditd package is not installed on the system.

○ system_auditing.auditd_installed: false

- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.3.10: Ensure use of privileged commands are collected

- **Status:** FAIL
- **Evidence:** The auditd package is not installed on the system.

○ system_auditing.auditd_installed: false

- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.3.11: Ensure events that modify /etc/group information are collected

- **Status:** FAIL
- **Evidence:** The auditd package is not installed on the system.

○ system_auditing.auditd_installed: false

- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.3.12: Ensure events that modify /etc/passwd information are collected

- **Status:** FAIL
- **Evidence:** The auditd package is not installed on the system.

○ system_auditing.auditd_installed: false

- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.3.13: Ensure events that modify /etc/shadow and /etc/gshadow are collected

- **Status:** FAIL
- **Evidence:** The auditd package is not installed on the system.
 - `system_auditing.auditd_installed: false`
- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.3.14: Ensure events that modify /etc/security/opasswd are collected

- **Status:** FAIL
- **Evidence:** The auditd package is not installed on the system.
 - `system_auditing.auditd_installed: false`
- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.3.15: Ensure events that modify /etc/nsswitch.conf file are collected

- **Status:** FAIL
- **Evidence:** The auditd package is not installed on the system.
 - `system_auditing.auditd_installed: false`
- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.3.16: Ensure events that modify /etc/pam.conf and /etc/pam.d/ information are collected

- **Status:** FAIL
- **Evidence:** The auditd package is not installed on the system.
 - `system_auditing.auditd_installed: false`
- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.3.17: Ensure unsuccessful file access attempts are collected

- **Status:** FAIL
- **Evidence:** The auditd package is not installed on the system.
 - `system_auditing.auditd_installed: false`
- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.3.18: Ensure discretionary access control permission modification events are collected

- **Status:** FAIL
- **Evidence:** The auditd package is not installed on the system.

○ system_auditing.auditd_installed: false

- **Recommendation:** Install auditd: sudo apt install auditd. Then configure and enable the service.

Rule 6.2.3.19: Ensure successful file system mounts are collected

- **Status:** FAIL
- **Evidence:** The auditd package is not installed on the system.

○ system_auditing.auditd_installed: false

- **Recommendation:** Install auditd: sudo apt install auditd. Then configure and enable the service.

Rule 6.2.3.20: Ensure session initiation information is collected

- **Status:** FAIL
- **Evidence:** The auditd package is not installed on the system.

○ system_auditing.auditd_installed: false

- **Recommendation:** Install auditd: sudo apt install auditd. Then configure and enable the service.

Rule 6.2.3.21: Ensure login and logout events are collected

- **Status:** FAIL
- **Evidence:** The auditd package is not installed on the system.

○ system_auditing.auditd_installed: false

- **Recommendation:** Install auditd: sudo apt install auditd. Then configure and enable the service.

Rule 6.2.3.22: Ensure file deletion events by users are collected

- **Status:** FAIL
- **Evidence:** The auditd package is not installed on the system.

○ system_auditing.auditd_installed: false

- **Recommendation:** Install auditd: sudo apt install auditd. Then configure and enable the service.

Rule 6.2.3.23: Ensure events that modify the system's Mandatory Access Controls are collected

- **Status:** FAIL
- **Evidence:** The auditd package is not installed on the system.

○ `system_auditing.auditd_installed: false`

- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.3.24: Ensure successful and unsuccessful attempts to use the `chcon` command are collected

- **Status:** FAIL
- **Evidence:** The `auditd` package is not installed on the system.

○ `system_auditing.auditd_installed: false`

- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.3.25: Ensure successful and unsuccessful attempts to use the `setfacl` command are collected

- **Status:** FAIL
- **Evidence:** The `auditd` package is not installed on the system.

○ `system_auditing.auditd_installed: false`

- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.3.26: Ensure successful and unsuccessful attempts to use the `chacl` command are collected

- **Status:** FAIL
- **Evidence:** The `auditd` package is not installed on the system.

○ `system_auditing.auditd_installed: false`

- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.3.27: Ensure successful and unsuccessful attempts to use the `usermod` command are collected

- **Status:** FAIL
- **Evidence:** The `auditd` package is not installed on the system.

○ `system_auditing.auditd_installed: false`

- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.3.28: Ensure kernel module loading unloading and modification is collected

- **Status:** FAIL
- **Evidence:** The auditd package is not installed on the system.

○ system_auditing.auditd_installed: false

- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.3.29: Ensure the audit configuration is immutable

- **Status:** FAIL
- **Evidence:** The auditd package is not installed on the system.

○ system_auditing.auditd_installed: false

- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.3.30: Ensure the running and on disk configuration is the same

- **Status:** INFORMATIONAL
- **Evidence:** The auditd package is not installed on the system.

○ system_auditing.auditd_installed: false

- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.4.1: Ensure the audit log file directory mode is configured

- **Status:** FAIL
- **Evidence:** The auditd package is not installed on the system.

○ system_auditing.auditd_installed: false

- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.4.2: Ensure audit log files mode is configured

- **Status:** FAIL
- **Evidence:** The auditd package is not installed on the system.

○ system_auditing.auditd_installed: false

- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.4.3: Ensure audit log files owner is configured

- **Status:** FAIL
- **Evidence:** The auditd package is not installed on the system.

○ system_auditing.auditd_installed: false

- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.4.4: Ensure audit log files group owner is configured

- **Status:** FAIL
- **Evidence:** The auditd package is not installed on the system.

○ `system_auditing.auditd_installed: false`

- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.4.5: Ensure audit configuration files mode is configured

- **Status:** FAIL
- **Evidence:** The auditd package is not installed on the system.

○ `system_auditing.auditd_installed: false`

- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.4.6: Ensure audit configuration files owner is configured

- **Status:** FAIL
- **Evidence:** The auditd package is not installed on the system.

○ `system_auditing.auditd_installed: false`

- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.4.7: Ensure audit configuration files group owner is configured

- **Status:** FAIL
- **Evidence:** The auditd package is not installed on the system.

○ `system_auditing.auditd_installed: false`

- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.4.8: Ensure audit tools mode is configured

- **Status:** FAIL
- **Evidence:** The auditd package is not installed on the system.

○ `system_auditing.auditd_installed: false`

- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.4.9: Ensure audit tools owner is configured

- **Status:** FAIL
- **Evidence:** The `auditd` package is not installed on the system.
 - `system_auditing.auditd_installed: false`
- **Recommendation:** Install `auditd`: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.4.10: Ensure audit tools group owner is configured

- **Status:** FAIL
- **Evidence:** The `auditd` package is not installed on the system.
 - `system_auditing.auditd_installed: false`
- **Recommendation:** Install `auditd`: `sudo apt install auditd`. Then configure and enable the service.

Section 6.3 – Integrity Checking (AIDE)

Rule 6.3.1: Ensure AIDE is installed

- **Status:** FAIL
- **Evidence:** The `aide` package is not installed on the system.
 - `aide_integrity_checking.aide_installed: false`
- **Recommendation:** Install AIDE: `sudo apt install aide aide-common`. Run `aideinit` to create the initial database.

Rule 6.3.2: Ensure filesystem integrity is regularly checked

- **Status:** FAIL
- **Evidence:** The `aide` package is not installed on the system.
 - `aide_integrity_checking.aide_installed: false`
- **Recommendation:** Install AIDE: `sudo apt install aide aide-common`. Then configure a cron job or systemd timer to regularly check filesystem integrity (e.g., `sudo aide --check`).

Rule 6.3.3: Ensure cryptographic mechanisms are used to protect the integrity of audit tools

- **Status:** FAIL
- **Evidence:** The `aide` package is not installed on the system.
 - `aide_integrity_checking.aide_installed: false`
- **Recommendation:** Install AIDE: `sudo apt install aide aide-common`. Configure AIDE to monitor critical audit tools and their configurations with strong cryptographic hashes (e.g., `sha512`).

Section 7.1 – Configure system file and directory access

Rule 7.1.1: Ensure access to /etc/passwd is configured

- **Status:** PASS
- **Evidence:** The /etc/passwd file has appropriate permissions (mode 0o644, owner root, group root).
 - `file_permissions.fixed_files[0].mode_octal: 0644`
 - `file_permissions.fixed_files[0].owner: root`
 - `file_permissions.fixed_files[0].group: root`
- **Recommendation:** None.

Rule 7.1.2: Ensure access to /etc/passwd- is configured

- **Status:** PASS
- **Evidence:** The /etc/passwd- file has appropriate permissions (mode 0o644, owner root, group root).
 - `file_permissions.fixed_files[1].mode_octal: 0644`
 - `file_permissions.fixed_files[1].owner: root`
 - `file_permissions.fixed_files[1].group: root`
- **Recommendation:** None.

Rule 7.1.3: Ensure access to /etc/group is configured

- **Status:** PASS
- **Evidence:** The /etc/group file has appropriate permissions (mode 0o644, owner root, group root).
 - `file_permissions.fixed_files[2].mode_octal: 0644`
 - `file_permissions.fixed_files[2].owner: root`
 - `file_permissions.fixed_files[2].group: root`
- **Recommendation:** None.

Rule 7.1.4: Ensure access to /etc/group- is configured

- **Status:** PASS
- **Evidence:** The /etc/group- file has appropriate permissions (mode 0o644, owner root, group root).
 - `file_permissions.fixed_files[3].mode_octal: 0644`
 - `file_permissions.fixed_files[3].owner: root`
 - `file_permissions.fixed_files[3].group: root`
- **Recommendation:** None.

Rule 7.1.5: Ensure access to /etc/shadow is configured

- **Status:** PASS

- **Evidence:** The `/etc/shadow` file has appropriate permissions (mode 0o640, owner root, group shadow).

- `file_permissions.fixed_files[4].mode_octal: 0640`
- `file_permissions.fixed_files[4].owner: root`
- `file_permissions.fixed_files[4].group: shadow`

- **Recommendation:** None.

Rule 7.1.6: Ensure access to `/etc/shadow-` is configured

- **Status:** PASS
- **Evidence:** The `/etc/shadow-` file has appropriate permissions (mode 0o640, owner root, group shadow).

- `file_permissions.fixed_files[5].mode_octal: 0640`
- `file_permissions.fixed_files[5].owner: root`
- `file_permissions.fixed_files[5].group: shadow`

- **Recommendation:** None.

Rule 7.1.7: Ensure access to `/etc/gshadow` is configured

- **Status:** PASS
- **Evidence:** The `/etc/gshadow` file has appropriate permissions (mode 0o640, owner root, group shadow).

- `file_permissions.fixed_files[6].mode_octal: 0640`
- `file_permissions.fixed_files[6].owner: root`
- `file_permissions.fixed_files[6].group: shadow`

- **Recommendation:** None.

Rule 7.1.8: Ensure access to `/etc/gshadow-` is configured

- **Status:** PASS
- **Evidence:** The `/etc/gshadow-` file has appropriate permissions (mode 0o640, owner root, group shadow).

- `file_permissions.fixed_files[7].mode_octal: 0640`
- `file_permissions.fixed_files[7].owner: root`
- `file_permissions.fixed_files[7].group: shadow`

- **Recommendation:** None.

Rule 7.1.9: Ensure access to `/etc/shells` is configured

- **Status:** PASS
- **Evidence:** The `/etc/shells` file has appropriate permissions (mode 0o644, owner root, group root).

- `file_permissions.fixed_files[8].mode_octal: 0644`
- `file_permissions.fixed_files[8].owner: root`
- `file_permissions.fixed_files[8].group: root`

- **Recommendation:** None.

Rule 7.1.10: Ensure access to /etc/security/opasswd is configured

- **Status:** PASS
- **Evidence:** The /etc/security/opasswd file has appropriate permissions (mode 0o600, owner root, group root).

- `file_permissions.fixed_files[9].mode_octal: 0600`
- `file_permissions.fixed_files[9].owner: root`
- `file_permissions.fixed_files[9].group: root`

- **Recommendation:** None.

Rule 7.1.11: Ensure world writable files and directories are secured

- **Status:** PASS
- **Evidence:** No world-writable files or directories without the sticky bit were found in the scanned mounts (/ and /boot).

- `file_permissions.world_writable.world_writable_files[0].paths: []`
- `file_permissions.world_writable.world_writable_dirs_no_sticky[0].paths: []`

- **Recommendation:** None.

Rule 7.1.12: Ensure no files or directories without an owner and a group exist

- **Status:** PASS
- **Evidence:** No unowned or ungrouped files or directories were found in the scanned mounts (/ and /boot).

- `file_permissions.unowned.unowned_or_ungrouped[0].paths: []`

- **Recommendation:** None.

Rule 7.1.13: Ensure SUID and SGID files are reviewed

- **Status:** INFORMATIONAL
- **Evidence:** The following SUID/SGID files were found on the system:

- `/usr/bin/umount (SUID root:root 4755)`
- `/usr/bin/chsh (SUID root:root 4755)`
- `/usr/bin/sudo (SUID root:root 4755)`
- `/usr/bin/su (SUID root:root 4755)`
- `/usr/bin/passwd (SUID root:root 4755)`
- `/usr/bin/ssh-agent (SGID root:_ssh 2755)`
- `/usr/bin/fusermount3 (SUID root:root 4755)`
- `/usr/bin/newgrp (SUID root:root 4755)`
- `/usr/bin/chage (SGID root:shadow 2755)`
- `/usr/bin/expiry (SGID root:shadow 2755)`

- /usr/bin/gpasswd (SUID root:root 4755)
- /usr/bin/crontab (SGID root:crontab 2755)
- /usr/bin/mount (SUID root:root 4755)
- /usr/bin/chfn (SUID root:root 4755)
- /usr/lib/dbus-1.0/dbus-daemon-launch-helper (SUID root:messagebus 4754)
- /usr/lib/x86_64-linux-gnu/utempter/utempter (SGID root:utmp 2755)
- /usr/lib/openssh/ssh-keysign (SUID root:root 4755)
- /usr/lib/polkit-1/polkit-agent-helper-1 (SUID root:root 4755)
- /usr/sbin/pam_extrausers_chkpwd (SGID root:shadow 2755)
- /usr/sbin/unix_chkpwd (SGID root:shadow 2755)

- **Recommendation:** Regularly review the list of SUID/SGID files to ensure that only authorized and necessary programs have these permissions. Remove SUID/SGID bits from any files that do not require them.

Section 7.2 – Local User and Group Settings

Rule 7.2.1: Ensure accounts in /etc/passwd use shadowed passwords

- **Status:** PASS
- **Evidence:** All user accounts listed in `user_chage_info` have password status indicating they use shadowed passwords (no unhashed passwords found).
- **Recommendation:** None.

Rule 7.2.2: Ensure /etc/shadow password fields are not empty

- **Status:** PASS
- **Evidence:** The root account's password status is *, indicating it's locked or uses a shadowed password. All other user accounts also show non-empty password fields (either L for locked or a valid hash).

```
○ user_accounts.root_and_system_accounts.root_shadow_password_status
  *
```

- **Recommendation:** None.

Rule 7.2.3: Ensure all groups in /etc/passwd exist in /etc/group

- **Status:** PASS
- **Evidence:** No orphan GIDs were found in /etc/passwd that do not exist in /etc/group.

```
○ user_accounts.local_users_and_groups_evidence.orphan_gids_in_passwd
  []
```

- **Recommendation:** None.

Rule 7.2.4: Ensure shadow group is empty

- **Status:** PASS
- **Evidence:** The shadow group has no members.

- `user_accounts.local_users_and_groups_evidence.shadow_group_memk`
`[]`

- **Recommendation:** None.

Rule 7.2.5: Ensure no duplicate UIDs exist

- **Status:** PASS
- **Evidence:** No duplicate UIDs were found on the system.

- `user_accounts.local_users_and_groups_evidence.duplicate_uids:`
`{}`

- **Recommendation:** None.

Rule 7.2.6: Ensure no duplicate GIDs exist

- **Status:** PASS
- **Evidence:** No duplicate GIDs were found on the system.

- `user_accounts.local_users_and_groups_evidence.duplicate_gids:`
`{}`

- **Recommendation:** None.

Rule 7.2.7: Ensure no duplicate user names exist

- **Status:** PASS
- **Evidence:** No duplicate usernames were found on the system.

- `user_accounts.local_users_and_groups_evidence.duplicate_usernam`
`{}`

- **Recommendation:** None.

Rule 7.2.8: Ensure no duplicate group names exist

- **Status:** PASS
- **Evidence:** No duplicate group names were found on the system.

- `user_accounts.local_users_and_groups_evidence.duplicate_groupna`
`{}`

- **Recommendation:** None.

Rule 7.2.9: Ensure local interactive user home directories are configured

- **Status:** PASS
- **Evidence:** No issues were found with local interactive user home directories (e.g., missing, incorrect permissions).

- `user_accounts.local_users_and_groups_evidence.home_directory_is`
`[]`

- **Recommendation:** None.

Rule 7.2.10: Ensure local interactive user dot files access is configured

- **Status:** PASS
- **Evidence:** No issues were found with local interactive user dot file permissions.

○ `user_accounts.local_users_and_groups_evidence.dotfile_permissions`
[]

- **Recommendation:** None.

SUID/SGID Triage Report

```
{ "findings": [], "summary": { "suspicious_found": 0, "message": "No unexpected SUID/SGID files detected." } }
```